VELOOP REWARDS — USER BEHAVIOR & ENGAGEMENT INTELLIGENCE REPORT

Data Analyst Internship · Task 02 · Product Analytics / User Behavior / Engagement Intelligence

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Observation window: 2026-04-01 – 2026-09-15 · Generated from the pipeline in this repository

Data authenticity statement. Every figure in this report is calculated from the fictional, simulated workbook *VELOOP_Simulated_User_Behavior_1000.xlsx* supplied for the internship. Nothing here describes actual VELOOP users, retention, engagement, acquisition or product performance. Wording follows the brief: “In the provided simulated dataset…”. The engagement score is an analyst-created model, not an official VELOOP metric. All experiments are proposals; none was run.

Statement classes used throughout

[DOCUMENTED PRODUCT FACT] Supported by official VELOOP documentation as cited in the brief

[SIMULATED DATA RESULT] Calculated from the simulated dataset

[ANALYST INTERPRETATION] What a result may indicate

[PRODUCT HYPOTHESIS] A proposed explanation or change to test

[TO BE VERIFIED] Requires confirmation from product/engineering

01. Executive Summary

SIMULATED DATA – NOT ACTUAL VELOOP REWARDS USER DATA. Every figure in this report is calculated from the fictional dataset supplied for this task. It shows how the analysis works and what it would find; it does not describe real VELOOP users, retention or performance.

Scope: 1,000 simulated users, 7,913 sessions and 35,704 feature events, 2026-04-01 – 2026-09-15. All metrics are recomputed from raw logs; every user count uses one shared definition (Appendix D reconciles numbers that look alike).

KPI (simulated)	Value	Numerator / denominator
Active users	984	of 1,000 registered
Activation (14-day)	58.8%	530 of 902 eligible users
Repeat users (≥2 active days)	83.9%	826 of 984 active users
D7 return (days 1–7)	47.3%	430 of 910 eligible users
D30 return (days 24–30)	41.8%	299 of 716 eligible users
Task completion	61.2%	5,627 of 9,198 finished task attempts

Most important findings

[SIMULATED DATA RESULT] Most simulated users take days, not minutes, to open a first session. Median time from registration to first session is 6.1 days; only 13.0% of eligible users complete a meaningful activity within 1 day and 30.2% within 3 days.

[SIMULATED DATA RESULT] Second-day return is the first major user-level drop-off. Among 716 users with a full 30-day horizon, 179 (25.0%) were active on only one day in their first 30 days.

[SIMULATED DATA RESULT] Return rates depend strongly on calendar period, not on who the users are. D7 return rises from 26.3% for the 2026-04 first-active cohort to 56.4% for 2026-08; the daily active share of the registered base rose from 5.09% (May) to 11.34% (last 30 days).

[SIMULATED DATA RESULT] Fast activators return more, even after controlling for join month. Users activating within 7 days have a 26.4pp higher fourth-week (D30) return than slower activators; within registration months the gap is 15.3pp.

[SIMULATED DATA RESULT] Two in five resolved tasks end in abandonment or failure, with no dominant failure reason. 3,571 of 9,198 resolved task records (38.8%) were abandoned (22.3%) or failed (16.5%).

Top actions and how to test them

Action	Test	Primary metric: baseline → target	Sample size
Reduce task abandonment and failure	EXP-03 (proposed A/B test)	Task completion rate (resolved): 61.2% → 64.2%	4,078 per arm
Shorten registration → first activity	EXP-01 (proposed A/B test)	14-day activation rate: 58.8% → 63.8%	1,489 per arm
Create a reason to return on day 2	EXP-02 (proposed A/B test)	D1 return rate: 14.5% → 17.5%	2,344 per arm

These are associations in simulated data, not proven causes. Each action should be validated by its experiment (Section 24) on production traffic before roll-out. Source code, the anonymised dataset and an independent verification script are provided (Appendix E).

02. Dataset Validation

Dataset Validation & Data Quality Impact

Ten sheets were profiled before any analysis. 03_Feature_Events is the primary behavioural source; 02_Sessions and 05_Task_Activity supply session windows and the task funnel; 04_Daily_Activity, 06_Milestones and 07_Feature_Usage are treated as validation references; 10_Event_Data_Quality_Audit is a QA copy with injected issues.

sheet	rows	columns	Null cells	Duplicate rows	Min datetime	Max datetime
01_Users_1000	1000	16	846	0	2026-04-01 02:00:41	2026-09-15 20:26:50
02_Sessions	7913	12	466	0	2026-04-02 20:14:28	2026-09-15 22:14:12
03_Feature_Events	35704	13	0	0	2026-04-02 20:14:31	2026-09-15 22:13:50
04_Daily_Activity	6667	17	0	0	2026-04-02 00:00:00	2026-09-15 00:00:00
05_Task_Activity	9846	12	10494	0	2026-04-02 20:14:41	2026-09-15 21:59:48
06_Milestones	4799	6	40	0	2026-04-02 08:18:00	2026-09-15 22:13:28
07_Feature_Usage	15	7	0	0	NaT	NaT
08_Data_Dictionary	26	7	0	0	NaT	NaT
09_README_Status	15	2	0	0	NaT	NaT
10_Event_Data_Quality_Audit	35714	13	0	10	2026-04-02 20:14:31	2026-09-15 22:13:50

Classification summary of 73 checks: VALID 29, WARNING 18, INVALID 14, INVESTIGATE 12.

Issue id	dataset	check	count	pct	classification	severity	Action taken
DQ-004	Users	Personal fields present (name, email, age, gender, city, state, referral_code)	7	43.75	WARNING	Medium	Dropped from every processed/analytical/dashboard output; raw workbook git-ignored.
DQ-005	Users	PDF-specified user fields absent from workbook	11	57.89	INVESTIGATE	Medium	Derived from sessions/events where possible; level/XP balance marked 'Not supported by available dataset'.
DQ-006	Users	Registered users with no session	16	1.6	WARNING	Low	Retained; analysed as zero-activity users.
DQ-007	Users	platform has a single value ('Web') for all users	1000	100	WARNING	Medium	Device (primary_device) used as the platform dimension; platform flagged as non-discriminating.
DQ-008	Users	onboarding_status='Not Started' but user has feature events	150	98.04	INVESTIGATE	Low	Reported descriptively only.
DQ-009	Users	account_status='Suspended' users with activity	65	98.48	INVESTIGATE	Low	Retained; status used descriptively only.
DQ-014	Sessions	Extremely long sessions (> Q3+3·IQR = 176s)	82	1.04	WARNING	Low	Retained and flagged (is_long_session); listed in anomaly register.
DQ-015	Sessions	Zero-event sessions (no rows in 03_Feature_Events)	302	3.82	WARNING	Medium	Retained as sessions (login proxy); excluded from per-session behaviour metrics ('valid sessions').
DQ-016	Sessions	events_in_session ≠ raw event count	7537	95.25	INVALID	High	Field not used; events/session recomputed from 03_Feature_Events.
DQ-017	Sessions	features_used_count ≠ raw distinct features	6750	85.3	INVALID	Medium	Field not used; recomputed from raw events.
DQ-018	Sessions	Overlapping sessions for the same user	38	0.48	WARNING	Low	Retained and flagged (is_overlapping); impact tested in data-quality impact analysis.
DQ-019	Sessions	exit_feature null	466	5.89	WARNING	Low	Exit analysis reported from both the field and the last raw event.
DQ-020	Sessions	exit_feature ≠ feature of last raw event (event-backed sessions)	6674	93.19	INVESTIGATE	Medium	Primary exit = last raw event feature (timestamp-ordered); field reported for comparison.
DQ-021	Sessions	session_status has a single value ('Completed')	7913	100	WARNING	Low	Field not used analytically.
DQ-022	Sessions	Task completed after its parent session ended	5601	99.54	INVESTIGATE	High	Session duration interpreted as 'logged session window', not true time-in-app.
DQ-031	Events	Repeated identical events (all fields equal except event_id)	17	0.05	WARNING	Low	Second+ copies excluded from the clean layer (flagged, not deleted from raw).

Issue id	dataset	check	count	pct	classification	severity	Action taken
DQ-032	Events	Same user + timestamp + event_name (any status/session)	140	0.39	WARNING	Low	Flagged in anomaly register.
DQ-033	Events	Duplicate sequence_number within a session	10789	30.22	INVALID	High	Event order derived from event_datetime (tie-break event_id).
DQ-034	Events	sequence_number not monotonic with timestamp (sessions with >2 events, |rho|<0.5)	3183	58.24	INVALID	High	sequence_number ignored for journey analysis.
DQ-035	Events	Completion event with no matching start event in the same session	4183	83.53	INVESTIGATE	High	Feature funnels computed at user level (ever exposed / started / completed).
DQ-036	Events	Completion timestamp earlier than start timestamp (same session)	423	51.27	INVALID	Medium	Flagged (sequence_anomaly); no time-to-complete metric derived from events.
DQ-037	Events	Multiple events with identical timestamp in one session	3384	9.48	WARNING	Low	Ties broken by event_id.
DQ-038	Events	streak_broken logged with event_status='Success'	919	2.57	WARNING	Medium	Stage taxonomy overrides raw status: streak_broken classified as 'negative'.
DQ-040	Events	PDF example events absent from simulated taxonomy	13	35.14	INVESTIGATE	High	Proxies documented; missing events added to tracking-gap register.
DQ-049	Tasks	Pending tasks older than 7 days at data end	534	82.41	WARNING	Medium	Pending excluded from completion-rate denominator; reported separately.
DQ-050	Milestones	first_login ≠ user's first session_start	388	99.74	INVALID	High	Lifecycle milestones re-derived from raw sessions/events.
DQ-051	Milestones	third_active_day recorded for users with < 3 active days	82	19.71	INVALID	High	Milestone counts re-derived from raw activity.
DQ-052	Milestones	seventh_active_day recorded for users with < 7 active days	188	44.66	INVALID	High	Milestone counts re-derived from raw activity.
DQ-053	Milestones	Milestones for users with no sessions	40	0.83	INVALID	Medium	Excluded (milestone sheet not used as a source of truth).
DQ-054	Milestones	source_event belongs to a different feature than the milestone (e.g. first_spin ← wallet_viewed)	1809	92.2	INVALID	Medium	Not used.
DQ-055	Milestones	Missing session_id	40	0.83	WARNING	Low	Not used.
DQ-056	Milestones	milestone_value semantics undocumented (uniform 1–100 across all types)	4799	100	INVESTIGATE	Low	Not used.
DQ-058	Daily Activity	session_duration_minutes ≠ sum of session durations (±0.1 min)	6559	98.45	INVALID	High	Field not used; time metrics come from 02_Sessions.
DQ-059	Daily Activity	session_count ≠ sessions started that day	57	0.86	WARNING	Low	Session counts from 02_Sessions.
DQ-060	Daily Activity	User-days with sessions but no daily row	249	3.6	WARNING	Low	Active day defined as a day with ≥1 feature event.
DQ-061	Daily Activity	active_flag has a single value ('Yes')	6667	100	WARNING	Low	Not used.
DQ-062	Daily Activity	Record volume below PDF recommendation (15,000–30,000)	6667	44.45	INVESTIGATE	Low	Documented as source discrepancy.
DQ-065	Feature Usage	Sheet is a 15-row aggregate (PDF suggests 8,000–15,000 feature-usage records)	15	0.19	INVESTIGATE	Low	Derived user×feature table built in clean layer.
DQ-066	Event Audit	Rows in 10_Event_Data_Quality_Audit vs 03_Feature_Events	10	0.03	WARNING	Low	03_Feature_Events used as primary source; audit sheet used to test cleaning rules.
DQ-067	Event Audit	Duplicated event rows (same event_id, identical content)	10	0.03	INVALID	Low	Deduplicated on event_id in impact test.
DQ-068	Event Audit	Non-canonical feature names (lower-case variants)	5	0.01	INVALID	Low	Normalised via case-insensitive mapping in impact test.
DQ-069	Event Audit	event_status differs from 03 (e.g. 'success ' with case/whitespace drift)	4	0.01	INVALID	Low	Normalised (strip + title-case) in impact test.
DQ-070	README / PDF	Project duration: README says '22–25 day', PDF says '20–24 Days'	1	100	INVESTIGATE	Low	PDF treated as the formal requirement; flagged.
DQ-072	README / PDF	Recommended volumes vs actual (Sessions 8–15k, Daily 15–30k, Feature Usage 8–15k)	3	42.86	INVESTIGATE	Low	Analysis proceeds on supplied data.

Full register (including VALID checks, denominators and analytical impact): outputs/tables/data_quality_report.csv / .xlsx.

[TO BE VERIFIED] Source discrepancy requiring confirmation: the workbook README describes a '22–25 day' project while the PDF states '20–24 Days'; sheet volumes (sessions, daily activity, feature usage) are below the PDF's recommended ranges.

03. User Lifecycle Analysis

Meaningful activity (definition). A meaningful activity is the earliest point at which a simulated user completes an action that the product is designed to reward. Views and starts show exposure or intent only; badges and level-ups are consequences awarded by the system rather than user actions. Included: ad_completed, captcha_completed, daily_reward_claimed, follow_task_completed, mining_activity, mission_completed, referral_created, referral_shared, spin_completed, streak_claimed, tap_completed; plus completed tasks. Excluded: views/opens, starts, skips/failures/abandons, badge_earned, level_up, mining_stopped, streak_broken. Limitation: Event-level completions cannot be tied to a preceding start within the same session (see DQ-035); 'meaningful' therefore means 'a completion was logged', not 'a full flow was observed'. Value of each action (points earned) is not in the dataset.

Two counts are shown for every stage: **cumulative** = users who met this stage and every earlier one; **standalone** = users who meet this stage's rule regardless of earlier stages. Stages with the same name have identical counts in the lifecycle, activation funnel, engagement funnel and dashboard.

Stage	Definition	Users (cumulative)	% of registered	% of previous	Users (standalone)
Registration	User record in 01_Users_1000	1000	100	100	1000
First login	≥1 session (login proxied by session existence)	984	98.4	98.4	984
First feature interaction	≥1 feature event of any kind	984	98.4	100	984
First meaningful activity	≥1 meaningful completion (completion event or completed task)	984	98.4	100	984
Repeated activity (2nd active day)	Active on at least 2 distinct calendar days	826	82.6	83.9	826
Multi-feature usage	Meaningful completions in ≥8 distinct features (P75 of active users)	235	23.5	28.5	265
Regular engagement	Active in ≥3 distinct weeks after the first active day	211	21.1	89.8	615

metric	< 1 hour	1–24 hours	1–3 days	3–7 days	7–14 days	14–30 days	> 30 days	No activity
Registration → first session	20	141	192	167	138	152	174	16
Registration → first feature event	19	135	196	165	140	153	176	16
Registration → first meaningful activity	19	116	181	162	150	169	187	16

milestone	Users derived from raw	Users in milestone sheet	difference	Pct of registered derived
first_login	984	389	-595	98.4
first_feature_use	984	423	-561	98.4
first_task_start	984	393	-591	98.4
first_task_completion	981	403	-578	98.1
first_reward_activity	984	392	-592	98.4
third_active_day	735	416	-319	73.5
seventh_active_day	424	421	-3	42.4
first_streak	533	376	-157	53.3
first_mission	771	394	-377	77.1
first_spin	489	387	-102	48.9
first_referral_activity	811	406	-405	81.1
level_up	615	399	-216	61.5

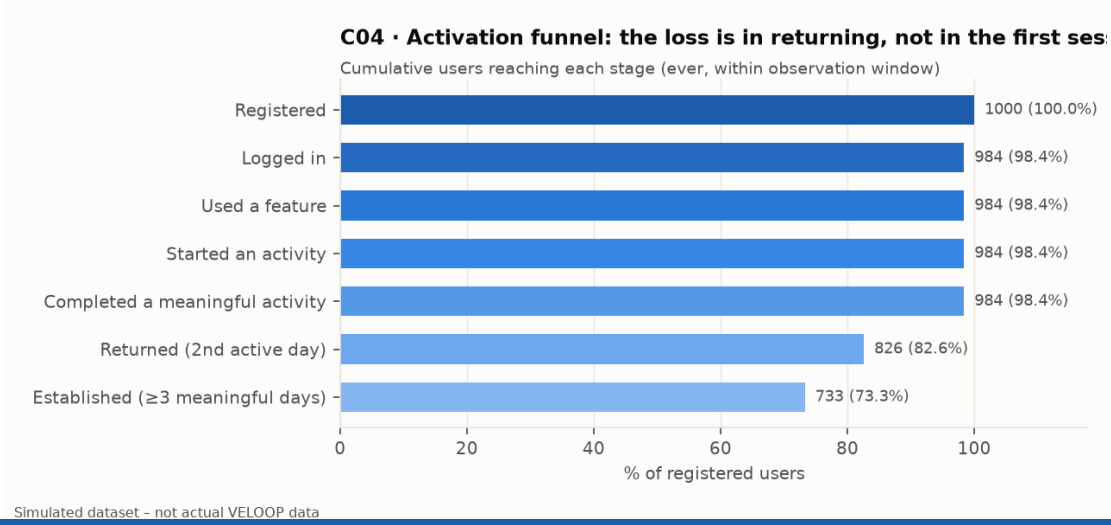
[ANALYST INTERPRETATION] The lifecycle is front-loaded: the first session reliably produces a completion, while repeat and regular engagement are where the cohort thins out.

04. Activation Analysis

Definition. First meaningful activity within 14 days of registration; denominator = users registered at least 14 days before data end (2026-09-15). **Window choice.** Selection rule: the smallest tested window that captures the majority (≥50%) of users who activate within 60 days. Capture by window: 1d 6.2%, 3d 19.4%, 7d 35.9%, 14d 56.3%, 30d 81.0%. Short windows misclassify most eventual activators because the simulated cohort takes a median of 6.1 days to open a first session. The gain per extra day falls from 2.91pp (days 8–14) to 1.54pp (days 15–30), so 14 days is the elbow; longer windows also shrink the eligible population (902 users eligible at 14d vs 816 at 30d). The 7-day rate (45.2%) is reported alongside as an 'early activation' view.

Window days	Eligible users	Activated users	Activation rate %	Capture of 60d activators %	Gain per extra day (pp)	selected
1	994	129	13	6.2		False

Window days	Eligible users	Activated users	Activation rate %	Capture of 60d activators %	Gain per extra day (pp)	selected
3	980	296	30.2	19.4	6.6	False
7	953	431	45.2	35.9	4.12	False
14	902	530	58.8	56.3	2.91	True
30	816	614	75.2	81	1.54	False



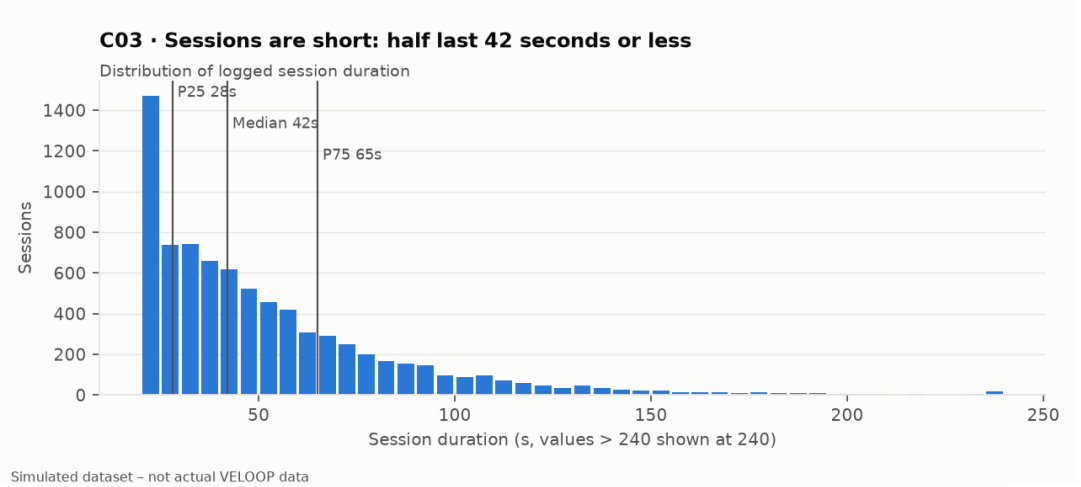
Stage	Definition	Users (cumulative)	% of 1,000 registered	% of previous	Users (standalone)	Source	Limitation
Registered	User record in 01_Users_1000	1000	100	100	1000	01_Users_1000	Registration time only; no pre-registration exposure.
Logged in	≥1 session (login proxied by session existence)	984	98.4	98.4	984	02_Sessions	No login event exists in the simulated taxonomy.
Used a feature	≥1 feature event of any kind	984	98.4	100	984	03_Feature_Events	Includes passive views.
Started an activity	≥1 start event (ad/captcha/spin/mission/tap/mining) or a task start	984	98.4	100	984	03 + 05	Daily Reward, Streak, Follow & Referral have no start event.
Completed a meaningful activity	≥1 meaningful completion (completion event or completed task)	984	98.4	100	984	03 + 05	Completion cannot be linked to a specific start (DQ-035).
Returned (2nd active day)	Active on at least 2 distinct calendar days	826	82.6	83.9	826	03_Feature_Events	Right-censored: recent users have had less time to return.
Established (≥3 meaningful days)	Meaningful activity on at least 3 distinct days	733	73.3	88.7	733	03 + 05	Analyst threshold.

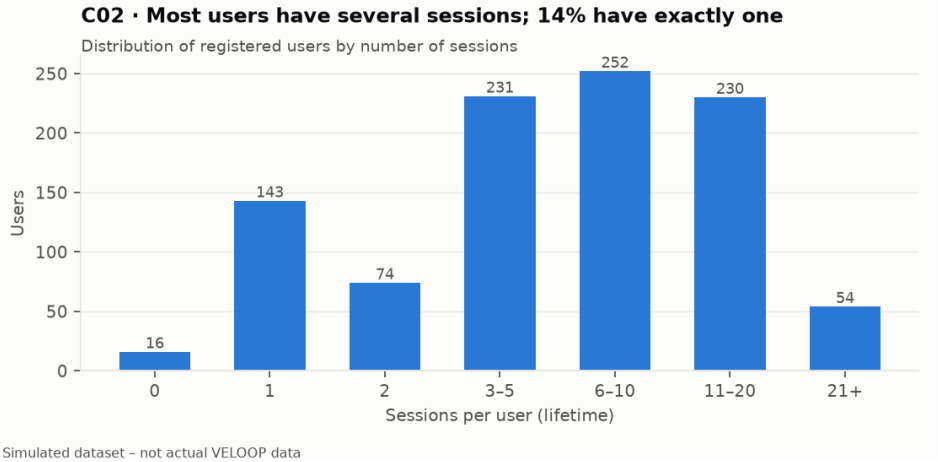
05. Session Behavior Analysis

Metric	Value
Total sessions	7913
Valid (event-backed) sessions	7611
Sessions / registered user	7.913
Sessions / active user	8.042
Average duration (s)	51.81
Median duration (s)	42.0
P25 / P75 / P90 / P95 (s)	28 / 65 / 94 / 117
Events / valid session	4.689
Features / valid session	3.733
One-session users	143
Repeat-session users	841
High-frequency users (≥11 sessions, P75)	284

Duration bands: Quartile-based on observed durations: Short ≤ 28s (P25), Medium ≤ 65s (P75), Long > 65s, Extreme > 176s (Q3 + 3·IQR).

Duration band	sessions	Pct sessions	Mean raw events	Mean meaningful	Meaningful session share	Meaningful per minute
Short (≤P25)	2065	26.1	4.39	1.26	0.65	3.35
Medium (P25–P75]	3915	49.5	4.42	1.32	0.67	1.8
Long (>P75)	1851	23.4	4.84	1.35	0.67	0.87
Extreme (>Q3+3·IQR)	82	1	4.2	1.21	0.62	0.33





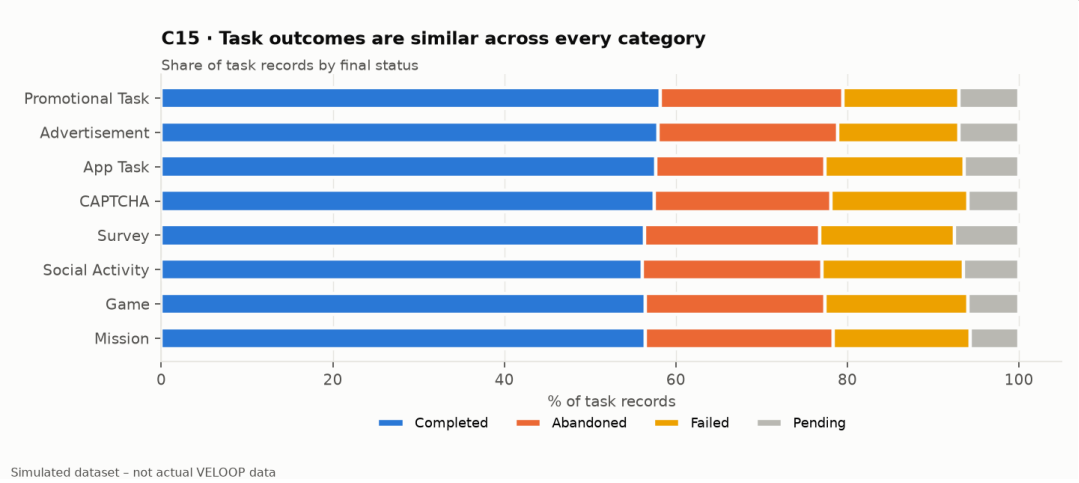
[ANALYST INTERPRETATION] Longer sessions are not deeper (ρ=0.027); short sessions deliver the most meaningful actions per minute. Duration should not be treated as 'good engagement'.

06. Feature Usage Analysis

Feature	Users	% active	Events	Sessions	First-feature users	Repeat users	Avg uses	Median uses	Completion %	Abandon %	Repeat %
Watch Ads	836	85	3126	2412	66	544	3.74	3	49.7	50.3	65.1
Daily Reward	835	84.9	2498	1963	86	506	2.99	3			60.6
Streak Rewards	832	84.6	2734	2157	80	529	3.29	3			63.6
CAPTCHA	824	83.7	2731	2120	83	496	3.31	3	50.3	49.7	60.2
Referral	811	82.4	2521	2000	77	485	3.11	2			59.8
Wallet	810	82.3	2166	1736	70	464	2.67	2			57.3
Tap & Earn	800	81.3	2651	2051	89	495	3.31	3			61.9
Mining Rewards	799	81.2	2616	2085	67	486	3.27	3			60.8
Level System	773	78.6	2426	1944	55	467	3.14	3			60.4
Missions	771	78.4	2780	2176	69	517	3.61	3	50.1	49.9	67.1
Spin the Wheel	763	77.5	2424	1909	63	465	3.18	2			60.9
Exchange Center	720	73.2	1689	1428	46	363	2.35	2			50.4
Leaderboard	686	69.7	1835	1516	47	359	2.67	2			52.3
Follow & Earn	685	69.6	1659	1377	42	356	2.42	2			52
Badges	674	68.5	1831	1538	44	349	2.72	2			51.8

Completion/abandonment rates exist only for features that log an explicit negative outcome (Watch Ads, CAPTCHA, Missions). Task categories (05_Task_Activity):

Task category	records	users	Completion rate %	Abandonment rate %	Failure rate %	Pending share %	Median start to complete (s)
Mission	985	634	59.8	23.3	16.9	5.7	292.5
Game	1008	623	60	22.3	17.7	6	284
Social Activity	1020	636	60	22.4	17.6	6.5	296
Survey	1259	711	60.9	22.2	16.9	7.5	303
CAPTCHA	1143	674	61.1	21.9	17	5.9	316
App Task	1327	719	61.6	21.1	17.3	6.4	305
Advertisement	2203	882	62.3	22.6	15.1	7	309
Promotional Task	901	581	62.5	22.9	14.6	7	323

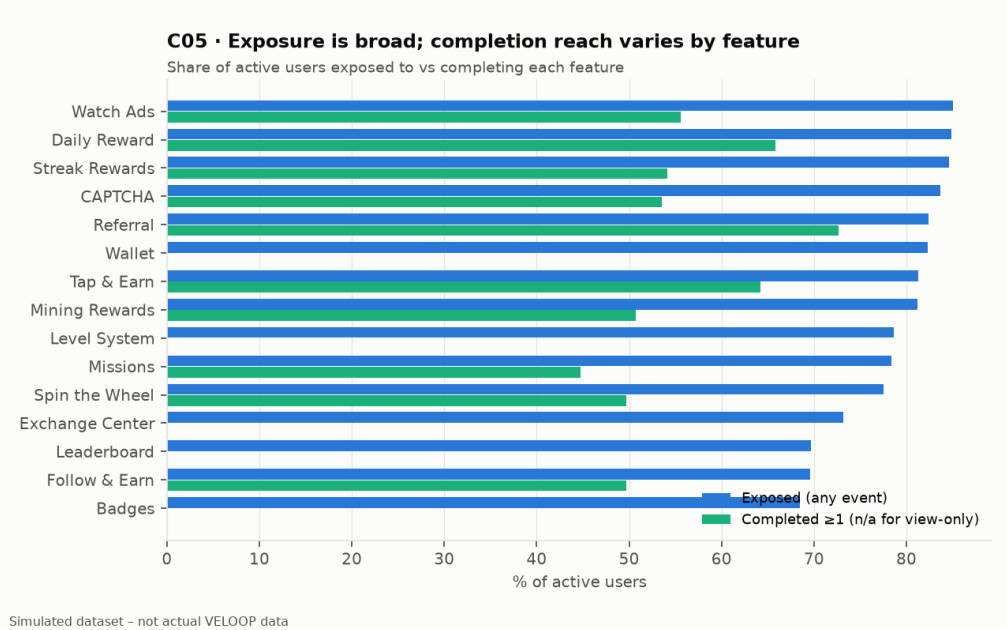


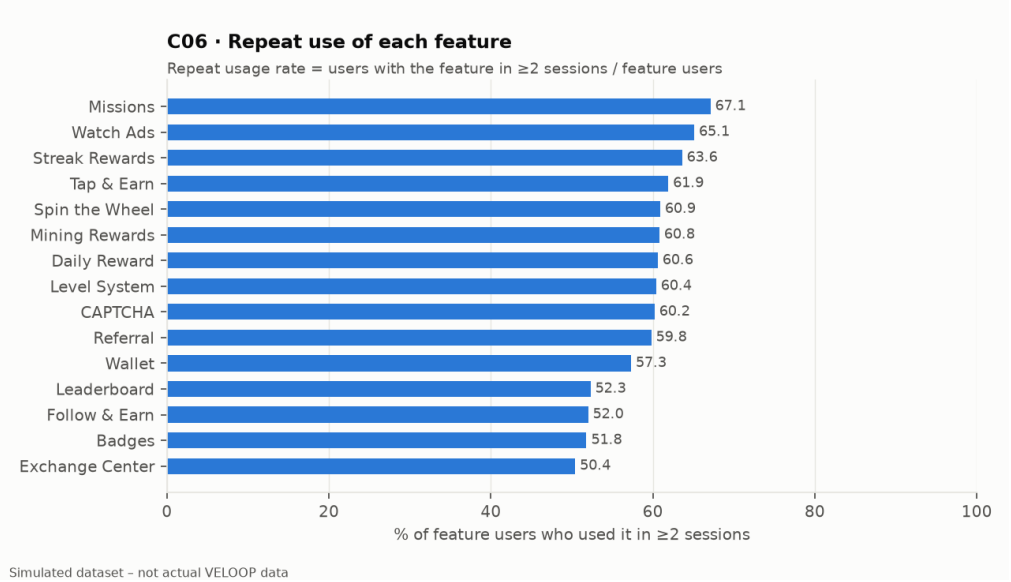
Simulated dataset – not actual VELOOP data

07. Feature Adoption & Repeat Usage

Exposure (any event) → Adoption (interaction beyond viewing) → Completion (≥1 'complete' event) → Repeat adoption (completed on ≥2 days) → Depth (completions per completer). View-only features have no adoption/completion stage.

feature	Exposure users	Adoption users	Completion users	Completion user rate %	Repeat completion users	Repeat completion rate %	Depth mean completions	Depth median completions
Watch Ads	836	836	547	65.4	236	43.1	1.93	2
Daily Reward	835	648	648	77.6	284	43.8	1.98	2
Streak Rewards	832	533	533	64.1	190	35.6	1.73	1
CAPTCHA	824	824	527	64	175	33.2	1.7	1
Referral	811	715	715	88.2	363	50.8	2.38	2
Wallet	810							
Tap & Earn	800	800	632	79	272	43	2.1	2
Mining Rewards	799	694	499	62.5	169	33.9	1.64	1
Level System	773	0	0	0	0			
Missions	771	716	440	57.1	142	32.3	1.6	1
Spin the Wheel	763	668	489	64.1	163	33.3	1.69	1
Exchange Center	720							
Leaderboard	686							
Follow & Earn	685	489	489	71.4	170	34.8	1.72	1
Badges	674	0	0	0	0			

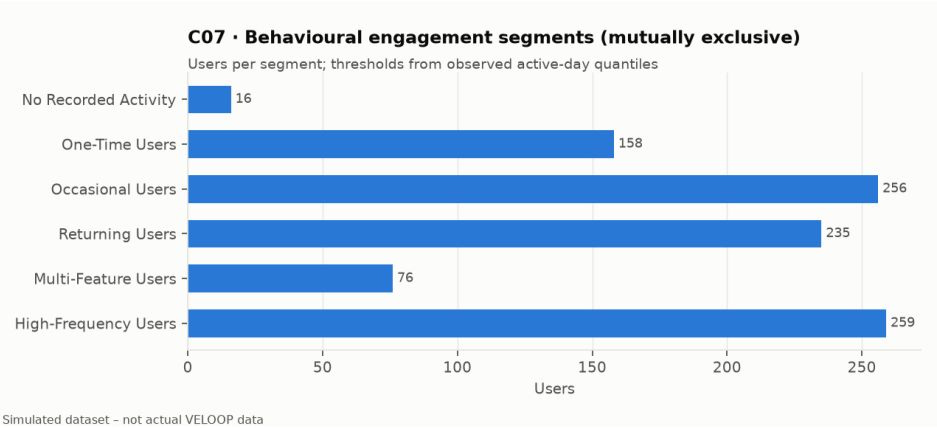



C06 · Repeat use of each feature
Repeat usage rate = users with the feature in ≥2 sessions / feature users
Missions
Watch Ads
Streak Rewards
Tap & Earn
Spin the Wheel
Mining Rewards
Daily Reward
Level System
CAPTCHA
Referral
Wallet
Leaderboard
Follow & Earn
Badges
Exchange Center
67.1
65.1
63.6
61.9
60.9
60.8
60.6
60.4
60.2
59.8
57.3
52.3
52.0
51.8
50.4
0
20
40
60
80
100
% of feature users who used it in ≥2 sessions
Simulated dataset – not actual VELOOP data

08. Engagement Segmentation

segment	rule	users	Pct users	Mean active days	Mean sessions	Mean features completed	Task completion rate %	rationale
No Recorded Activity	0 active days (registered, no feature events)	16	1.6	0	0	0		Separates registration-only accounts so they do not dilute behavioural averages.
One-Time Users	Exactly 1 active day	158	15.8	1	1.3	4.37	59.9	A single active day is the clearest 'did not return' signal (brief, segment A).
Occasional Users	2–4 active days and breadth below 8 completed features	256	25.6	2.99	3.64	3.67	61.7	Below the median active-user frequency (5 days).
Returning Users	5–9 active days and breadth below 8 completed features	235	23.5	6.71	7.84	5.06	62.3	At or above the median but below the top quartile (10 days).
Multi-Feature Users	2–9 active days and meaningful completions in ≥8 distinct features	76	7.6	6.43	8.84	8.57	61	Top-quartile breadth (P75 = 8 features) without top-quartile frequency – broad explorers who are not yet frequent.
High-Frequency Users	≥10 active days (top quartile of active users)	259	25.9	14.2	16.44	7.91	60.4	Top quartile of active days (P75 = 10).

Segments are mutually exclusive (hierarchical rules). Feature breadth correlates with frequency (Spearman ρ = 0.584), so 'Multi-Feature Users' captures broad explorers who are not yet high-frequency.



09. Engagement Score

Analyst-Created Simulated Engagement Score. Created for this internship analysis only. It is not an official VELOOP metric and is not used by VELOOP.

component	weight	transformation	reason
Activity Frequency (active days)	0.25	log1p + min-max (P99 cap)	Return on separate days is the most direct signal of repeated engagement in the brief.
Session Frequency (sessions)	0.1	log1p + min-max (P99 cap)	Adds visit intensity, but is ~0.95 rank-correlated with active days, so weighted low to avoid double counting.
Active-Day Consistency (active weeks / observable weeks, floor 4)	0.15	ratio, capped at 1	Rewards spread-out activity over bursts; partially corrects for tenure differences.
Feature Diversity (distinct features with a completion / 11)	0.15	ratio	Breadth of value actions; capped by the 11 completable simulated features.
Meaningful Completion (meaningful events + completed tasks)	0.25	log1p + min-max (P99 cap)	Completed value actions are what the product rewards; views alone are excluded.
Milestone Progression (level_up + badge_earned events)	0.1	log1p + min-max (P99 cap)	Progression is a consequence of activity, so weighted low to limit redundancy.

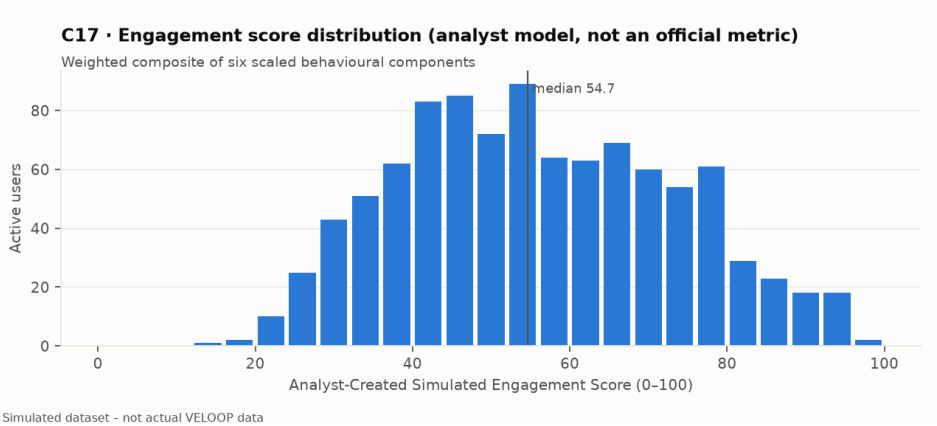
scenario	Spearman vs base	Top decile overlap %	Bottom decile overlap %
Equal weights	1	90.8	91.8
Drop Activity Frequency	0.98	88.8	83.7
Drop Session Frequency	1	96.9	91.8
Drop Active-Day Consistency	0.99	84.7	94.9
Drop Feature Diversity	0.99	89.8	87.8
Drop Meaningful Completion	0.99	90.8	83.7

scenario	Spearman vs base	Top decile overlap %	Bottom decile overlap %
Drop Milestone Progression	0.99	89.8	88.8
PCA (PC1 loadings) weights	1	93.9	93.9
Percentile-rank transform	0.99	89.8	92.9

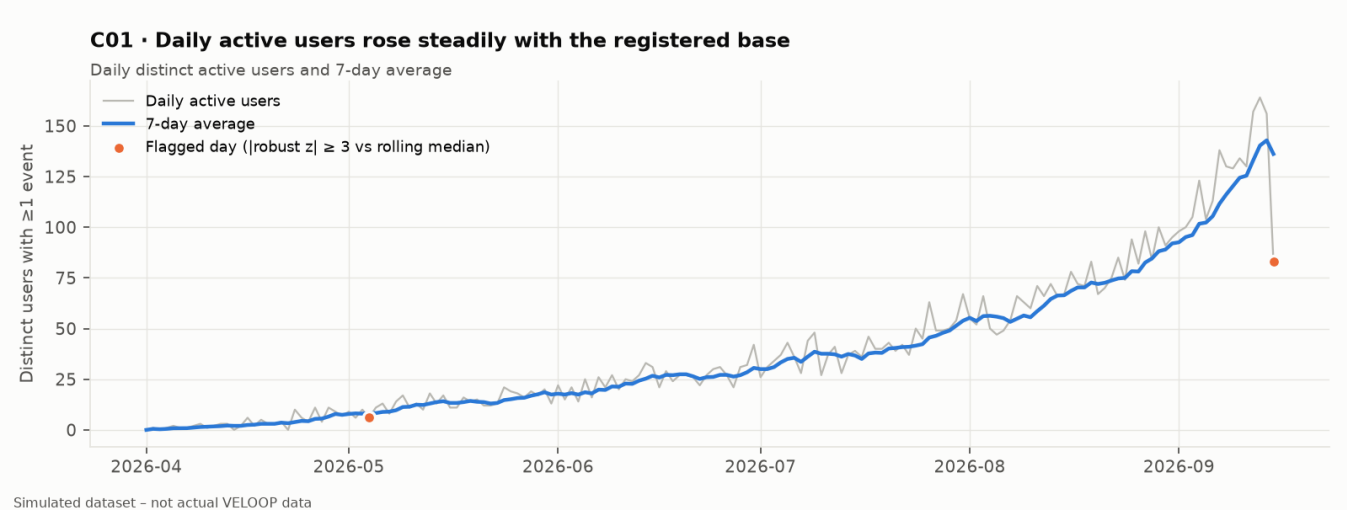
Distribution (active users): min 14.5, median 54.7, max 96.8. Predictive check: Score recomputed using only each user's first 14 days after first activity (users with ≥60 observable days); compared with active days on days 15–59. Spearman ρ = 0.326 (n = 499).

segment	count	mean	median	min	max
No Recorded Activity	16	0	0	0	0
One-Time Users	158	35.91	34.38	14.5	57.72
Occasional Users	256	42.56	42.7	23.26	60.24
Returning Users	235	57.24	56.69	36.29	76.27
Multi-Feature Users	76	66.31	66.42	50.05	79.75
High-Frequency Users	259	77.38	77.27	54.74	96.8

- Weights are analyst judgement; sensitivity tests show how much rankings depend on them.
- Components are correlated (frequency, sessions and completions move together), so the score is close to a single 'activity volume' dimension.
- Tenure bias: users registered recently have had less time to accumulate activity; the consistency floor only partially corrects this.
- No reward value, XP balance or level is available, so 'progression' uses counts of progression events.
- The score describes simulated behaviour; it has no validated link to any business outcome.



10. Daily & Time-Based Behavior



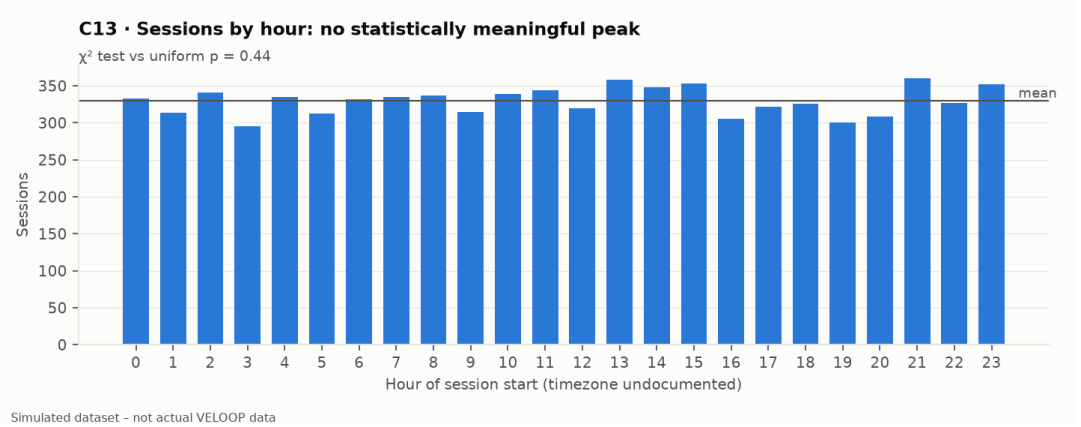
[SIMULATED DATA RESULT] DAU tracks the cumulative registered base (ρ = 0.987); the daily active share of registered users rose from 5.09% (May) to 11.34% (last 30 days). 2 day(s) flagged by robust-z detection.

date	dow	dau	Dau 7d avg	registrations	events	Robust z	flag
2026-05-04 00:00:00	Monday	6	8.29	4	30	-3.02	Drop
2026-09-15 00:00:00	Tuesday	83	136.14	6	644	-3.07	Drop

[ANALYST INTERPRETATION] Flags are statistical only. No campaign, release or notification calendar is available, so no flagged day is attributed to a cause.

dow	Avg dau	Avg sessions	Dau index vs 7d mean	Events per active user day	Meaningful share %	Task completion rate %
Monday	42.08	52.08	0.97	5.89	29.34	60.94
Tuesday	39.62	48.5	0.93	5.51	28.36	59.6
Wednesday	38.39	43.7	0.97	5.08	28.34	60.61
Thursday	37.67	43.04	1.06	4.96	29.25	60.85
Friday	38.83	46.08	0.94	5.22	29.08	61.56
Saturday	39.71	46.62	0.98	5.4	29.89	63.23
Sunday	43.08	51.5	1.11	5.34	29.07	61.47

daypart	sessions	Pct sessions	Sessions per hour of bucket	events	Avg session duration (s)	Task completion rate %
Morning (05-11)	2315	29.26	330.71	10508	52.49	61.15
Afternoon (12-16)	1684	21.28	336.8	7719	51.5	60.77
Evening (17-20)	1257	15.89	314.25	5366	52.12	60.93
Night (21-04)	2657	33.58	332.12	12094	51.28	61.58



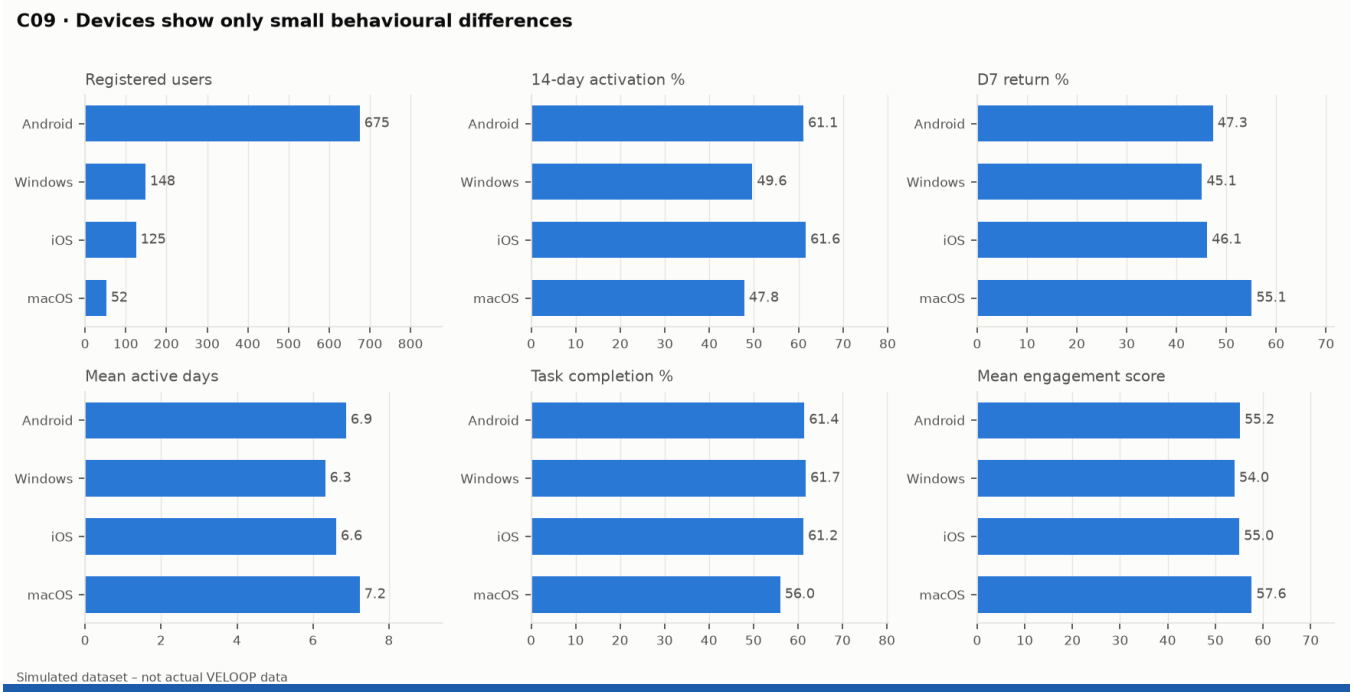
New vs returning activity (session level: first active day vs later days)

Activity type	sessions	users	Mean events	Mean features	Mean meaningful	Median duration (s)	Task completion rate %	Ad skip rate %	Mission abandon rate %
New (first active day)	1141	984	7.64	4.7	2.26	38	61.17	51.13	50.15
Returning (later days)	6470	825	4.17	3.56	1.2	43	61.25	50	49.81

[PRODUCT HYPOTHESIS] Because hour/day patterns are flat in the simulated data, it cannot inform notification timing, campaign scheduling or maintenance windows; those decisions need production data.

11. Device Analysis

platform = 'Web' for 100% of users, sessions and events; device (Android / iOS / Windows / macOS) is the only usable platform dimension. Device is constant per user in this dataset.

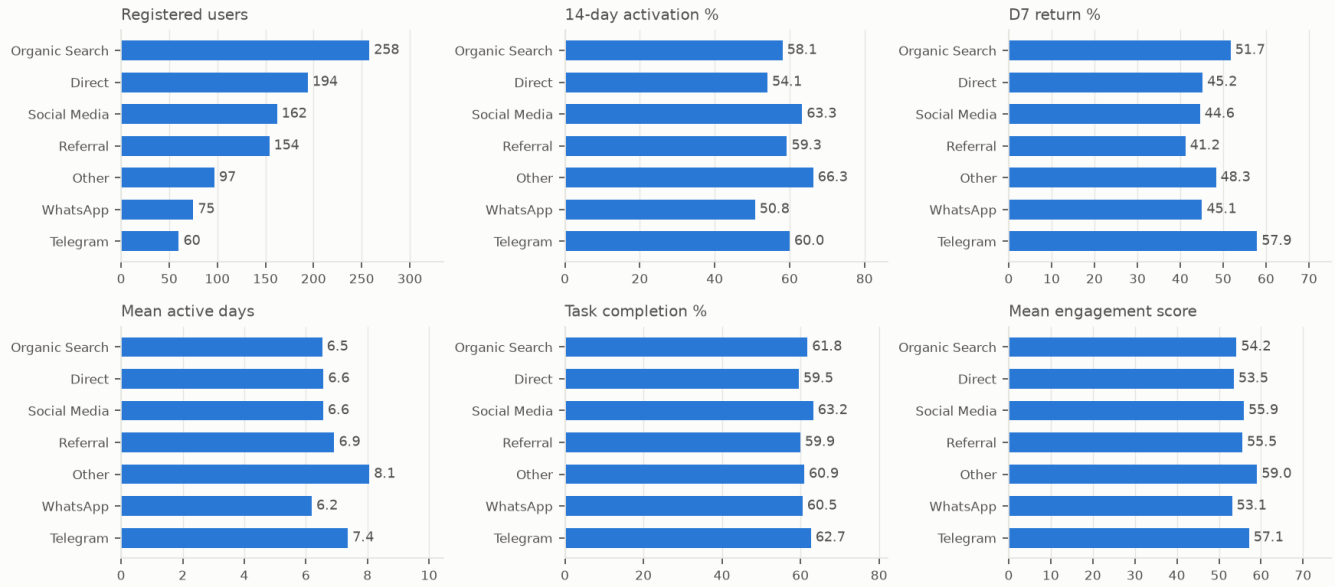


metric	test	P value	Effect size	Effect measure	interpretation
active days (active users)	Kruskal-Wallis	0.81	0	epsilon²	No statistically detectable difference at α=0.05
sessions (active users)	Kruskal-Wallis	0.84	0	epsilon²	No statistically detectable difference at α=0.05
feature diversity (active users)	Kruskal-Wallis	0.64	0	epsilon²	No statistically detectable difference at α=0.05
engagement score (active users)	Kruskal-Wallis	0.89	0	epsilon²	No statistically detectable difference at α=0.05
session duration (sessions)	Kruskal-Wallis	0.54	0	epsilon²	No statistically detectable difference at α=0.05
events per valid session	Kruskal-Wallis	0.01	0	epsilon²	Statistically detectable but negligible effect size
hours to first activity	Kruskal-Wallis	0.08	0	epsilon²	No statistically detectable difference at α=0.05
activation	Chi-square	0.03	0.1	Cramér's V	Statistically detectable but negligible effect size
D7 return	Chi-square	0.68	0.04	Cramér's V	No statistically detectable difference at α=0.05
task completion (resolved tasks)	Chi-square	0.13	0.02	Cramér's V	No statistically detectable difference at α=0.05

[ANALYST INTERPRETATION] Observed behavioural differences between devices are small. Possible explanations for the minor activation gap (e.g. desktop users registering with lower intent) are hypotheses only.

12. Acquisition Behavior Analysis

C08 · Acquisition sources differ in volume more than in behaviour



Simulated dataset – not actual VELOOP data

Acquisition source	users	Activation rate %	Median days to first activity	D7 return %	D30 return %	Mean active days	Mean feature diversity	Task completion rate %	Mean engagement score
Organic Search	258	58.08	6.5	51.72	41.99	6.53	11.74	61.76	54.16
Direct	194	54.07	7.35	45.2	37.41	6.57	11.75	59.54	53.51
Social Media	162	63.33	5.68	44.59	44.63	6.56	12.02	63.22	55.9
Referral	154	59.29	5.98	41.18	48.11	6.92	11.68	59.93	55.54
Other	97	66.29	5.69	48.31	43.66	8.06	12.29	60.93	59.04
WhatsApp	75	50.75	7.43	45.07	33.93	6.19	11.16	60.51	53.11
Telegram	60	60	5.64	57.89	38.1	7.36	12.07	62.66	57.13

[ANALYST INTERPRETATION] Organic Search brings the most registrations, but no source shows materially deeper behaviour; ranking sources by volume would be misleading. Simulated sources do not reflect real marketing performance.

13. User Journey Analysis

Order uses event_datetime (sequence_number is invalid). Consecutive events of the same feature are collapsed.

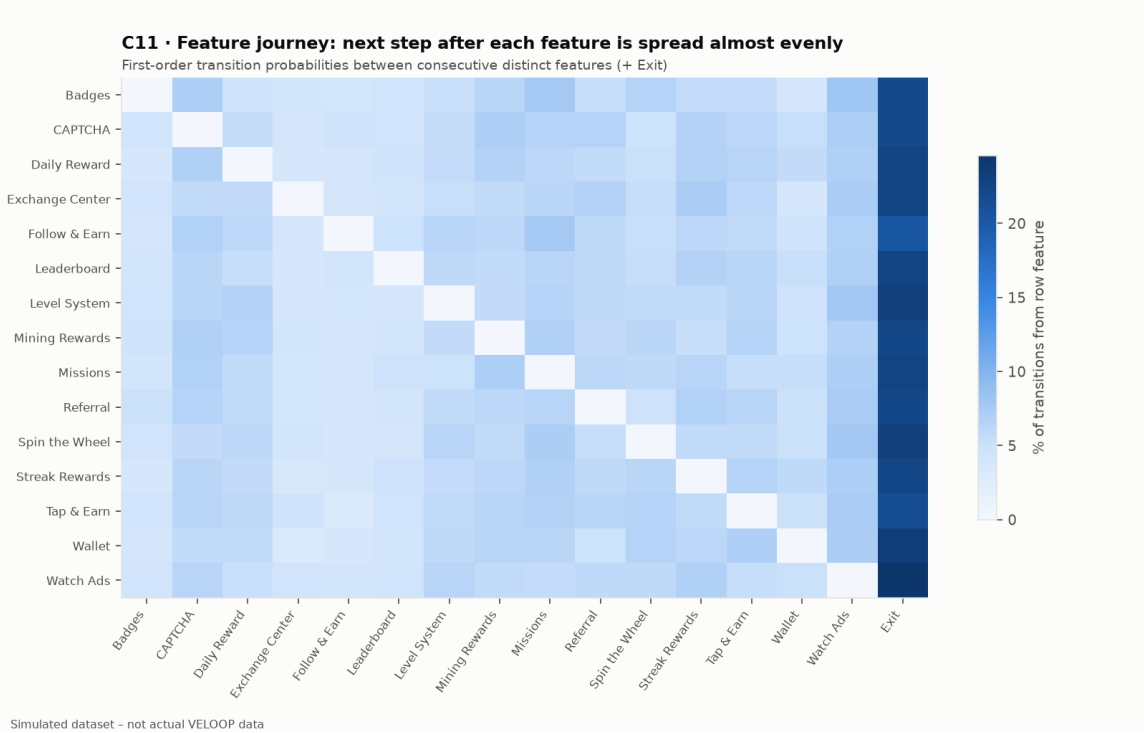
feature	First feature users	First pct of active	Second feature users
Tap & Earn	89	9	66
Daily Reward	86	8.7	71
CAPTCHA	83	8.4	81
Streak Rewards	80	8.1	81
Referral	77	7.8	78
Wallet	70	7.1	67
Missions	69	7	66
Mining Rewards	67	6.8	67

journey	sessions	Pct valid sessions
Session start → Watch Ads → Exit	89	1.17
Session start → Streak Rewards → Exit	75	0.99
Session start → Missions → Exit	69	0.91
Session start → CAPTCHA → Exit	62	0.81
Session start → Spin the Wheel → Exit	61	0.8
Session start → Referral → Exit	61	0.8
Session start → Tap & Earn → Exit	60	0.79
Session start → Daily Reward → Exit	57	0.75

type	journey	sessions
Successful (≥1 meaningful completion)	Session start → Referral → Exit	45
Successful (≥1 meaningful completion)	Session start → Watch Ads → Exit	35
Successful (≥1 meaningful completion)	Session start → Daily Reward → Exit	33
Successful (≥1 meaningful completion)	Session start → Streak Rewards → Exit	31
Successful (≥1 meaningful completion)	Session start → Tap & Earn → Exit	30
Successful (≥1 meaningful completion)	Session start → Spin the Wheel → Exit	19
Successful (≥1 meaningful completion)	Session start → Missions → Exit	17
Successful (≥1 meaningful completion)	Session start → CAPTCHA → Exit	17
Successful (≥1 meaningful completion)	Session start → Follow & Earn → Exit	17
Successful (≥1 meaningful completion)	Session start → Mining Rewards → Exit	14
Ends on abandon/fail/skip event	Session start → Watch Ads → Exit	29
Ends on abandon/fail/skip event	Session start → CAPTCHA → Exit	18
Ends on abandon/fail/skip event	Session start → Missions → Exit	13
Ends on abandon/fail/skip event	Session start → Level System → Watch Ads → Exit	7
Ends on abandon/fail/skip event	Session start → Daily Reward → Watch Ads → Exit	6
Ends on abandon/fail/skip event	Session start → Streak Rewards → CAPTCHA → Exit	6
Ends on abandon/fail/skip event	Session start → Streak Rewards → Missions → Exit	5
Ends on abandon/fail/skip event	Session start → CAPTCHA → Watch Ads → Exit	5
Ends on abandon/fail/skip event	Session start → Missions → Watch Ads → Exit	4
Ends on abandon/fail/skip event	Session start → CAPTCHA → Missions → Exit	4

feature	Last raw event feature	Last raw %	Exit feature field	Field %
Watch Ads	712	9.4	524	7
Missions	587	7.7	473	6.4
Streak Rewards	578	7.6	509	6.8
CAPTCHA	562	7.4	486	6.5

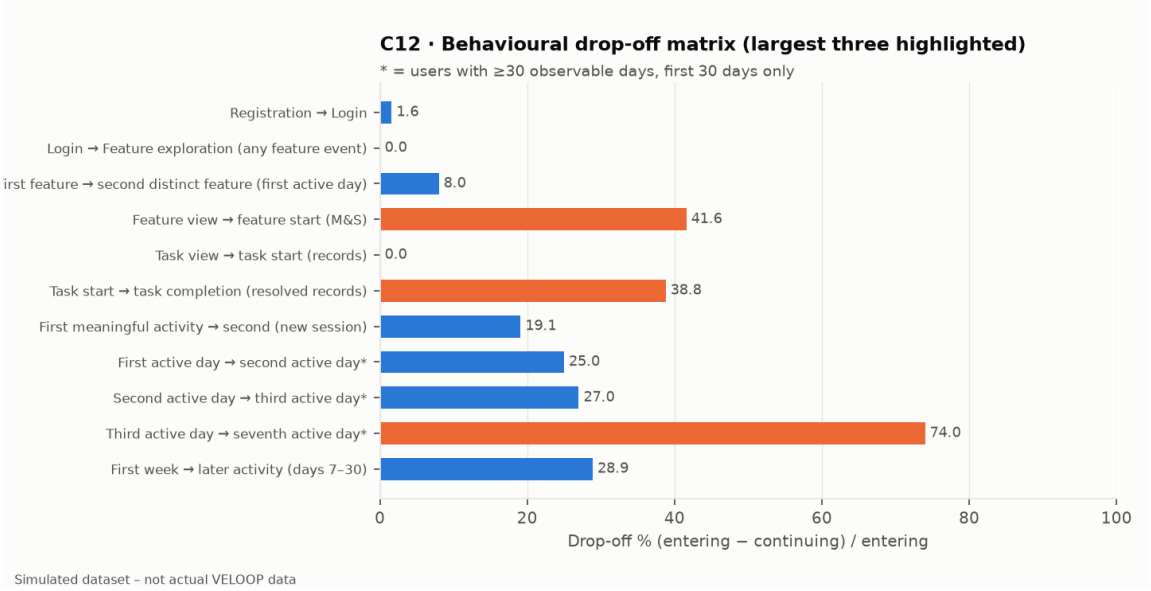
feature	Last raw event feature	Last raw %	Exit feature field	Field %
Mining Rewards	553	7.3	492	6.6
Tap & Earn	536	7	499	6.7
Level System	534	7	452	6.1
Spin the Wheel	531	7	459	6.2



[ANALYST INTERPRETATION] No path dominates (top path = 1.17% of sessions); transition lifts 0.79–1.27. Sequence frequency does not imply causation.

14. Drop-Off Analysis

stage	unit	entering	continuing	Drop off	Drop off %	observation	limitation	Product implication
Registration → Login	users	1000	984	16	1.6	16 simulated users never opened a session; median time to first session for the rest was 6.1 days.	No login/registration events; session existence is the login proxy.	Investigate the post-registration handoff (verification, first-open prompt) and the multi-day delay to first session.
Login → Feature exploration (any feature event)	users	984	984	0	0	Every user with a session also has feature events, but 302 individual sessions contain no events.	Zero-event sessions may be logging gaps rather than behaviour.	Instrument session_started/feature_impression to separate empty visits from tracking loss.
First feature → second distinct feature (first active day)	users	984	905	79	8	Median distinct features on the first active day: 4.	Breadth on day 1 depends on how many events a session contains.	Cross-feature discovery on day 1 appears high; low-breadth day-1 users are the target for discovery tests.
Feature view → feature start (user×feature, Missions & Spin)	user×feature pairs	901	526	375	41.6	Many viewers never record a start of the same feature; many starters never record a view (369 user×feature pairs).	Only 2 features log both a view and a start; views and starts are not linked within sessions (DQ-035).	Treat as a hypothesis signal: test clearer calls-to-action on mission and spin surfaces.
Task view → task start (records)	task records	9846	9846	0	0	Every task record has a start timestamp – view→start drop-off is not observable.	Task records appear to be created only when a task is started (no pure 'viewed' records).	Add task_viewed events independent of starts to measure task-list friction.
Task start → task completion (resolved records)	task records	9198	5627	3571	38.8	2051 abandoned and 1520 failed; 648 pending records excluded.	Pending tasks may later resolve; failure_reason is categorical only.	Largest record-level friction point – investigate the dominant failure reasons by category.
First meaningful activity → second meaningful activity (separate session)	users	965	781	184	19.1	Share of completers who complete again in another session.	Event-based completions only (task completions excluded here to keep a single source).	Post-completion prompts (next suggested activity) are a candidate investigation.
First active day → second active day (first 30 days)	users	716	537	179	25	Within their first 30 days, 179 of 716 users with a full 30-day horizon were active on only one day (all-time, unrestricted: 158 one-day users).	Restricted to users first active ≥30 days before data end; strongly calendar-dependent (see cohorts).	Second-day return is a primary user-level drop-off; post-first-day re-engagement is a key investigation area.
Second active day → third active day (first 30 days)	users	537	392	145	27	Share of returning users who return a second time within 30 days.	Same 30-day horizon population.	Habit-formation mechanics (streaks, daily rewards) are candidates for investigation here.
Third active day → seventh active day (first 30 days)	users	392	102	290	74	Progression to a 7th active day within the first 30 days.	Same 30-day horizon population.	Mid-term progression: missions/levels may matter more than onboarding here (hypothesis).
First week → later activity (days 7–30)	users	716	509	207	28.9	Users with ≥30 observable days who were active again between day 7 and day 30.	Restricted to users first active ≥30 days before data end.	Week-2+ re-engagement is the target for notification-timing experiments.



Engagement funnel (Work Package T)

Uses the same canonical stages as the activation funnel; the first six stages are identical. No 'viewed a feature' stage is used because Ads, CAPTCHA, Tap & Earn and Mining log no view events.

Stage	Definition	Users (cumulative)	% of 1,000 registered	% of previous	Users (standalone)	Limitation
Registered	User record in 01_Users_1000	1000	100	100	1000	Registration time only; no pre-registration exposure.
Logged in	≥1 session (login proxied by session existence)	984	98.4	98.4	984	No login event exists in the simulated taxonomy.
Used a feature	≥1 feature event of any kind	984	98.4	100	984	Includes passive views.
Started an activity	≥1 start event (ad/captcha/spin/mission/tap/mining) or a task start	984	98.4	100	984	Daily Reward, Streak, Follow & Referral have no start event.
Completed a meaningful activity	≥1 meaningful completion (completion event or completed task)	984	98.4	100	984	Completion cannot be linked to a specific start (DQ-035).
Returned (2nd active day)	Active on at least 2 distinct calendar days	826	82.6	83.9	826	Right-censored: recent users have had less time to return.
Multi-feature	Meaningful completions in ≥8 distinct features (P75 of active users)	235	23.5	28.5	265	Breadth rises with activity volume; not a strictly sequential step.
Regular (≥3 active weeks)	Active in ≥3 distinct weeks after the first active day	211	21.1	89.8	615	Users first active < 3 weeks before data end cannot qualify.

15. Repeat Engagement Analysis

metric	Multi-session users	Single-session users
users	841	143
mean_feature_types	12.09	10.17
mean_features_completed	5.79	4.56
mean_active_days	7.76	1.01
mean_sessions	9.24	1
mean_session_s	52.01	40.86
median_session_s	38	35
task_completion_rate_pct	61.39	59.91
daily_reward_claim_users_pct	67.78	54.55
streak_claim_users_pct	56.36	41.26
streak_broken_users_pct	57.67	41.96
mission_completed_users_pct	46.02	37.06
spin_completed_users_pct	51.72	37.76
level_up_users_pct	64.45	51.05
referral_action_users_pct	74.32	62.94
meaningful_per_session	1.23	8.1

First-active-day feature completion vs D7 return (volume-adjusted by first-day event tercile):

feature	Users completed day1	D7 return if completed %	D7 return if not %	Raw difference (pp)	Volume adjusted difference (pp)	Chi2 p value	reading
Streak Rewards	157	35.7	49.7	-14	1.6	0	Association detectable (not causal)
Watch Ads	153	36.6	49.4	-12.8	-1	0.01	Association detectable (not causal)
Mining Rewards	141	34	49.7	-15.6	-2.5	0	Association detectable (not causal)
Missions	111	32.4	49.3	-16.9	-4	0	Association detectable (not causal)
Tap & Earn	216	38.4	50	-11.6	-5.7	0	Association detectable (not causal)
CAPTCHA	155	32.9	50.2	-17.3	-6	0	Association detectable (not causal)
Daily Reward	210	33.3	51.4	-18.1	-7	0	Association detectable (not causal)
Referral	239	33.9	52	-18.1	-7.2	0	Association detectable (not causal)
Spin the Wheel	138	36.2	49.2	-13	-7.4	0.01	Association detectable (not causal)
Follow & Earn	133	30.1	50.2	-20.1	-7.5	0	Association detectable (not causal)

[ANALYST INTERPRETATION] The simulated dataset shows an association between spreading activity over visits and repeated engagement. It does not show that any feature causes return; raw feature differences mostly vanish after adjusting for first-day volume.

Streak & progression behaviour (Work Package Q)

- Users claiming streaks: 533 (54.2% of active); breaking: 545; both: 338.
- Maintaining (claims on consecutive days): 23 users (4.3%); longest run 5 days.
- Activity around first streak break: 0.83 active days in prior 7 vs 0.93 after (p = 0.23).

- Level-ups: 615 users (62.5%), 1266 events; 943 users touched progression features (Level, Badges, Leaderboard, Missions).
- Activity 14 days before vs after first level-up: 1.2 vs 1.34 active days (p = 0.22).
- Not supported by available dataset (fields absent from 01_Users_1000).

[DOCUMENTED PRODUCT FACT] Per the brief, VELOOP's level system describes XP as a progression mechanism based on valid activity and notes thresholds and benefits may change; simulated thresholds are therefore dataset-specific.

16. Cohort & Return Analysis

- cohort_date: First active date = calendar date of the user's first feature event (03_Feature_Events).
- secondary_cohort_date: Registration date (01_Users_1000.registration_datetime) – reported separately.
- active_definition: ≥1 feature event on that calendar day.
- denominator: Users whose entire window lies within the observation period (ends 2026-09-15); later users are excluded, not counted as non-returners.
- d30_supported: Yes – 716 users have a fully observable day 15–30 window.
- Windows: D1 = days 1–1 after cohort day 0; D7 = days 1–7 after cohort day 0; D14 = days 8–14 after cohort day 0; D30 = days 24–30 after cohort day 0

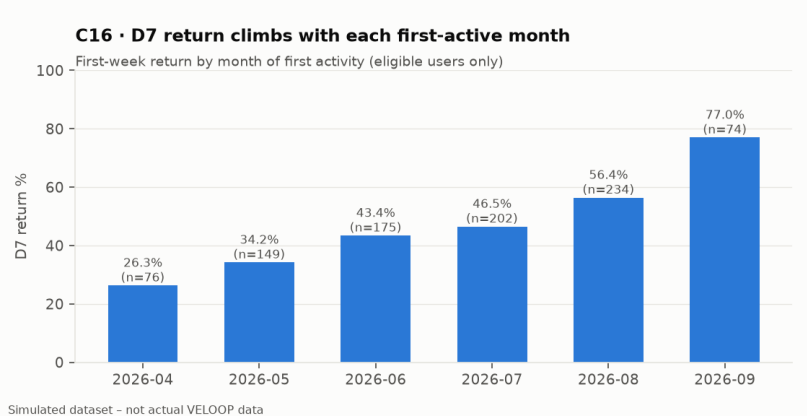
Primary retention table – one anchor (first active date), one active definition, eligible users only:

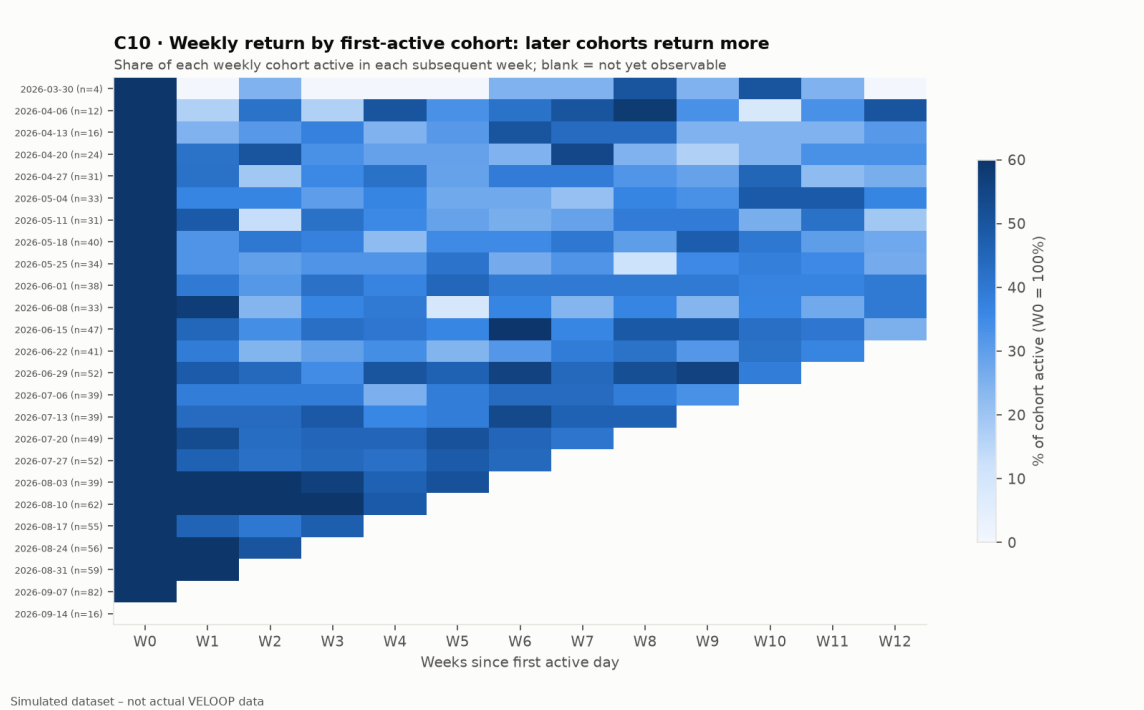
Window	Days after first active day	Eligible users	Returned users	Return rate %
D1	1	976	142	14.5
D7	1–7	910	430	47.3
D14	8–14	845	384	45.4
D30	24–30	716	299	41.8

D1 is a single day; D7, D14 and D30 are 7-day windows, so the rates are not cumulative and D30 can be close to D7. The registration-anchored comparison is in Appendix D.

Exact-day ('classic') return for comparison: day 1 14.5% (n=976), day 7 14.7% (n=910), day 14 9.6% (n=845), day 30 9.4% (n=716).

Cohort month	users	D1 rate %	D7 rate %	D14 rate %	D30 rate %	Mean active days
2026-04	76	2.6	26.3	28.9	32.9	9.43
2026-05	149	4	34.2	33.6	34.9	8.31
2026-06	175	5.1	43.4	43.4	37.7	7.99
2026-07	202	10.9	46.5	46.5	44.1	7.15
2026-08	234	17.9	56.4	58.1	58.8	5.76
2026-09	148	43.6	77	66.7		3.51





Behavioural cohorts on a fixed 30-day horizon

dimension	cohort	users	Active days 30d mean	Sessions 30d mean	Feature diversity 30d mean	Repeat activity %	Task completion rate %	D7 return %	D30 return %
Registration month	2026-04	175	2.35	2.37	7.83	61.7	59.4	25.1	29.1
Registration month	2026-05	160	2.66	2.75	8.07	65.6	61.5	32.5	30
Registration month	2026-06	166	3.39	3.64	9.12	79.5	61.5	45.8	36.1
Registration month	2026-07	151	4.84	5.38	10.39	84.8	63.1	57	58.3
Registration month	2026-08	64	7.73	9.27	12.25	100	61.2	73.4	81.2
Acquisition source	Direct	139	3.44	3.67	9.06	74.1	57.8	41	37.4
Acquisition source	Organic Search	181	3.72	4.03	9.19	76.8	61.6	47.5	42
Acquisition source	Other	71	4.38	4.93	9.75	81.7	59.5	45.1	43.7
Acquisition source	Referral	106	3.72	4.08	8.89	72.6	60.4	36.8	48.1
Acquisition source	Social Media	121	3.66	4.08	9.18	71.9	67.5	42.1	44.6
Acquisition source	Telegram	42	3.86	4.1	9.33	81	64.3	47.6	38.1
Acquisition source	WhatsApp	56	2.91	3.12	8.38	69.6	59.7	35.7	33.9
Device	Android	483	3.78	4.16	9.18	76	62.2	42.7	41.2
Device	Windows	107	3.42	3.68	8.93	73.8	61.9	44.9	42.1
Device	iOS	91	3.36	3.57	8.95	70.3	59.4	39.6	41.8
Device	macOS	35	3.57	3.8	9.31	77.1	56.1	42.9	48.6
First meaningful feature	CAPTCHA	75	4.15	4.53	9.43	78.7	58.1	41.3	46.7
First meaningful feature	Daily Reward	95	2.98	3.14	8.17	73.7	60.3	38.9	33.7
First meaningful feature	Follow & Earn	48	4.65	5.08	10.25	81.2	62.1	47.9	56.2
First meaningful feature	Mining Rewards	50	3.78	4.38	8.94	80	63.5	50	44
First meaningful feature	Missions	47	3.26	3.45	9.34	70.2	63.9	46.8	38.3
First meaningful feature	None (task only)	10	1.4	1.4	3.6	40	61.8	20	10
First meaningful feature	Referral	120	3.66	3.96	9.52	74.2	59	40	42.5
First meaningful feature	Spin the Wheel	43	3.7	3.88	9.58	79.1	65.8	44.2	41.9
First meaningful feature	Streak Rewards	70	3.79	4.1	9.47	75.7	64.6	40	37.1

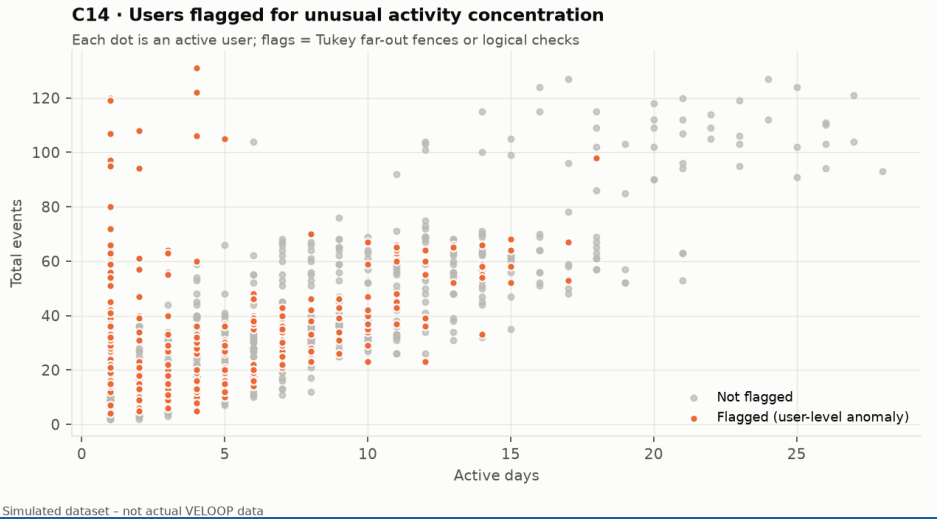
dimension	cohort	users	Active days 30d mean	Sessions 30d mean	Feature diversity 30d mean	Repeat activity %	Task completion rate %	D7 return %	D30 return %
First meaningful feature	Tap & Earn	96	3.83	4.34	8.98	72.9	60.9	43.8	43.8
First meaningful feature	Watch Ads	62	3.56	3.87	8.9	74.2	61.8	45.2	43.5
Activation timing	Within 24h	61	5.72	6.64	10.92	88.5	58.6	65.6	62.3
Activation timing	Day 2–7	247	4.76	5.32	9.84	88.3	62.9	55.5	55.5
Activation timing	After day 7	408	2.7	2.8	8.41	65	61	31.4	30.4

dimension	Kruskal p active days 30d	Chi2 p d30 return
Registration month	0	0
Acquisition source	0.36	0.54
Device	0.84	0.87
First meaningful feature	0.03	0.24
Activation timing	0	0

17. Behavioral Anomaly Analysis

Detect → quantify → document → flag for investigation. No fraud determination is made and no confidential anti-fraud rule is implied.

type	Entities flagged	population	Pct flagged	threshold	Recommended investigation
Extremely high event volume (user)	0	984	0	> 160 events (Q3+3·IQR)	Logging duplication / automation review.
Unusual activity concentration (events / active day)	84	984	8.54	> 19.3 events/day	Burst review.
High session event count	103	7611	1.35	> 18 events	Event-stream review.
High event density (events/min)	148	7611	1.94	> 27.6 events/min	Rate-pattern review.
Unusually long session	82	7913	1.04	> 176s	Idle-timeout review.
Repeated identical events	17	35704	0.05	> 1 identical copy	Client retry logic.
Excessive repeated event in a session	215	31605	0.68	> 3 of the same event	Idempotency review.
Repeated identical ≥4-event sequences	0	4213	0	≥2 sessions with same sequence	Session comparison.
Completion before start (same session)	423	825	51.27	completion ts < start ts	Clock/ordering pipeline.
Overlapping sessions	38	7913	0.48	start < previous end	Session-id logic.
Streak broken without any recorded claim	207	545	37.98	claims = 0	Streak state logging.
Task completion after session end	5601	5627	99.54	completed_at > session_end	Session definition.
Pending > 30 days	339	648	52.31	> 30 days pending	Validation SLA.



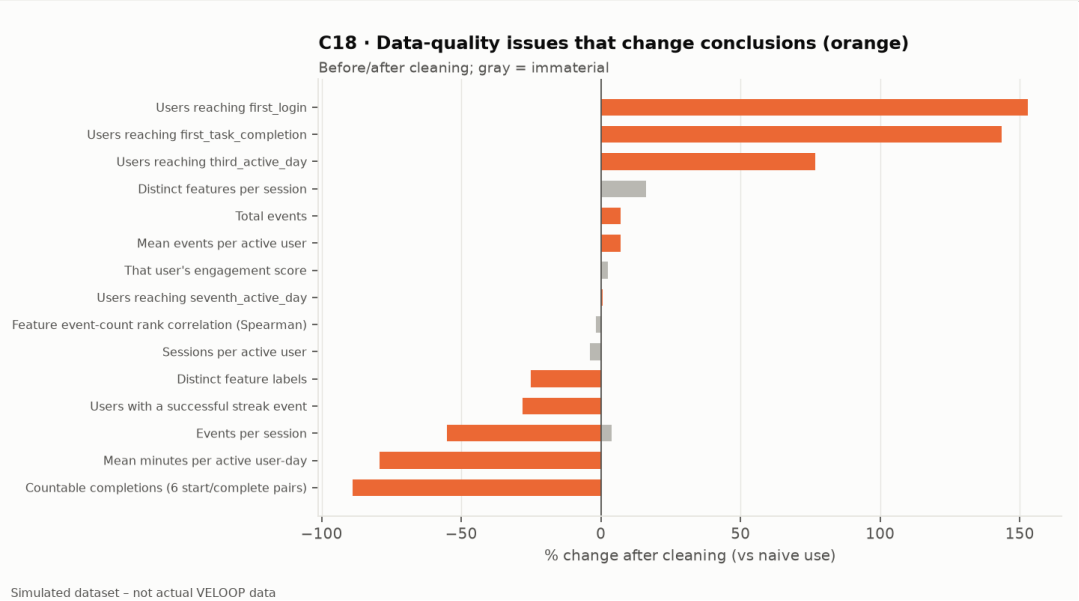
Anomaly id	type	entity	metric	Observed value	threshold	severity
ANM-0001	Excessive repeated feature events	SESS0000063	'badge_viewed' count in session	6	3	High
ANM-0002	Excessive repeated feature events	SESS0000063	'leaderboard_viewed' count in session	7	3	High
ANM-0003	Excessive repeated feature events	SESS0000262	'streak_claimed' count in session	5	3	High
ANM-0004	Excessive repeated feature events	SESS0000426	'wallet_viewed' count in session	5	3	High
ANM-0005	Excessive repeated feature events	SESS0000502	'exchange_viewed' count in session	5	3	High
ANM-0006	Excessive repeated feature events	SESS0001116	'leaderboard_viewed' count in session	8	3	High
ANM-0007	Excessive repeated feature events	SESS0001116	'level_up' count in session	5	3	High
ANM-0008	Excessive repeated feature events	SESS0001116	'level_viewed' count in session	5	3	High
ANM-0009	Excessive repeated feature events	SESS0001116	'spin_started' count in session	5	3	High
ANM-0010	Excessive repeated feature events	SESS0001116	'wallet_viewed' count in session	7	3	High

Full register: outputs/tables/anomaly_register.csv (1319 records).

18. Data Quality Impact

scenario	metric	Before cleaning	After cleaning	Absolute change	Percent change	Conclusion changes	explanation
Duplicate events (17 identical copies)	Total events	35,704	35,687	-17	-0.05	No	0.05% of events; immaterial.
Duplicate events (17 identical copies)	Events per active user	36.28	36.27	-0.02	-0.05	No	Immaterial at population level.
Duplicate events (17 identical copies)	Top-5 feature ranking identical (1 = yes)	1	1	0	0	No	Feature popularity ranking unchanged.
Audit sheet used as-is (duplicates, case variants)	Distinct feature labels	20	15	-5	-25	Yes (if uncorrected)	Lower-case variants create 5 phantom features.
Audit sheet used as-is (duplicates, case variants)	Event rows	35,714	35,704	-10	-0.03	No	10 duplicated rows double-count a handful of events.
Audit sheet used as-is (duplicates, case variants)	Events with status 'Success'	14,430	14,430	0	0	No	Offsetting errors: 4 duplicated 'Success' rows are added while the drifted statuses ('success ') drop out of exact-match filters – the totals agree only by coincidence.
Session-sheet counters (events_in_session)	Events per session	10.44	4.69	-5.75	-55.07	Yes	Field overstates session depth by >2×; session depth conclusions would be wrong.
Session-sheet counters (features_used_count)	Distinct features per session	3.21	3.73	0.52	16.1	Partly	Field understates breadth relative to raw events.
Zero-event sessions included vs excluded	Events per session	4.51	4.69	0.18	3.97	No	Small (~4%) dilution.
Zero-event sessions included vs excluded	Sessions per active user	8.04	7.74	-0.31	-3.82	No	Small dilution.
Overlapping sessions removed	Total sessions	7,913	7,875	-38	-0.48	No	0.5% of sessions.
Overlapping sessions removed	Median session duration (s)	42	42	0	0	No	Unchanged.
Milestone sheet vs raw derivation	Users reaching first_login	389	984	595	152.96	Yes	Milestone sheet is not consistent with raw activity.
Milestone sheet vs raw derivation	Users reaching first_task_completion	403	981	578	143.42	Yes	Milestone sheet is not consistent with raw activity.
Milestone sheet vs raw derivation	Users reaching third_active_day	416	735	319	76.68	Yes	Milestone sheet is not consistent with raw activity.
Milestone sheet vs raw derivation	Users reaching seventh_active_day	421	424	3	0.71	Yes	Milestone sheet is not consistent with raw activity.
Daily-activity duration field vs sessions	Mean minutes per active user-day	4.79	0.99	-3.81	-79.38	Yes	Daily field implies ~5 min/day vs ~1 min/day from session windows.
Require a preceding start in the same session (strict funnel)	Countable completions (6 start/complete pairs)	5,621	618	-5,003	-89.01	Yes	Strict sequence logic would discard most completions – conclusions about completion volume depend entirely on the linkage rule; user-level 'ever completed' logic is used instead.
streak_broken logged as 'Success'	Users with a successful streak event	740	533	-207	-27.97	Yes	Status-based success would count users who only broke a streak.
Simulated: 2% of events lose timestamps and are dropped	Mean active days per active user	6.78	6.77	-0.01	-0.12	No	Active-day metrics are robust to small random loss.
Simulated: 2% of events lose timestamps and are dropped	One-time users (1 active day)	158	158	0	0	No	Slight shift only.
Simulated: 2% of events lose user_id	Active users	984	984	0	0	No	Active-user counts barely move because users have many events.
Stress test: top user's events duplicated 20×	Total events	35,687	38,176	2,489	6.97	Yes (volume)	Event volume inflated by one user.
Stress test: top user's events duplicated 20×	Mean events per active user	36.27	38.8	2.53	6.97	Yes (mean)	Means are fragile to a single duplicated user.
Stress test: top user's events duplicated 20×	Median events per active user	31	31	0	0	No	Medians are robust – reason medians are reported alongside means.

scenario	metric	Before cleaning	After cleaning	Absolute change	Percent change	Conclusion changes	explanation
Stress test: top user's events duplicated 20×	Feature event-count rank correlation (Spearman)	1	0.98	-0.02	-1.79	Partly	Event-count popularity is distorted (top feature Watch Ads → Watch Ads); user-count based adoption metrics are unaffected.
Stress test: top user's events duplicated 20×	Unique users of top feature	836	836	0	0	No	Unique-user metrics are immune to duplication.
Stress test: top user's events duplicated 20×	That user's engagement score	72.79	74.59	1.8	2.47	No	P99 winsorisation caps count components, so duplication barely moves the score.
Stress test: top user's events duplicated 20×	Median engagement score (active users)	54.66	54.56	-0.1	-0.19	No	Population score robust.



19. Behavioral Findings

FND-01 · Most simulated users take days, not minutes, to open a first session

(Activation · confidence High)

[SIMULATED DATA RESULT] **Observation:** Median time from registration to first session is 6.1 days; only 13.0% of eligible users complete a meaningful activity within 1 day and 30.2% within 3 days.

[SIMULATED DATA RESULT] **Evidence:** 14-day activation = 530/902 eligible users (58.8%); 174 users waited > 30 days for a first session.

[ANALYST INTERPRETATION] **Interpretation:** The gap between sign-up and first use – not in-session friction – is the main activation delay in the simulated cohort.

Limitation: Registration timestamps carry no context (e.g. verification emails, app install) explaining the delay.

[PRODUCT HYPOTHESIS] **Product implication:** Investigate the post-registration handoff: verification flow, first-open prompts and welcome messaging.

FND-02 · Once a session happens, a completed activity almost always follows

(Activation · confidence Medium)

[SIMULATED DATA RESULT] **Observation:** 984 of 984 users with a session recorded a meaningful activity; 88.4% did so within 15 minutes of their first feature event.

[SIMULATED DATA RESULT] **Evidence:** Lifecycle: Login 984 → first meaningful 984.

[ANALYST INTERPRETATION] **Interpretation:** In-session activation is not the bottleneck in the simulated data; getting users to the first session and back again is.

Limitation: Completions are 'logged', not proven end-to-end flows (completion events lack linked starts, DQ-035).

[PRODUCT HYPOTHESIS] **Product implication:** Prioritise return and first-session-delay investigations over first-session UX changes.

FND-03 · Second-day return is the first major user-level drop-off **(Return · confidence High)**

[SIMULATED DATA RESULT] **Observation:** Among 716 users with a full 30-day horizon, 179 (25.0%) were active on only one day in their first 30 days.

[SIMULATED DATA RESULT] **Evidence:** All-time one-day users: 158 of 984 active users; D1 return = 14.5% (142/976).

[ANALYST INTERPRETATION] **Interpretation:** A quarter of users with sufficient observation never form a second visit.

Limitation: The dataset does not record why users stop or whether they received reminders.

[PRODUCT HYPOTHESIS] **Product implication:** Investigate post-first-day re-engagement (reminders, next-day reward previews).

FND-04 · Progression from 3 to 7 active days is the steepest drop in the active-day ladder **(Return · confidence Medium)**

[SIMULATED DATA RESULT] **Observation:** Within the first 30 days, 102 of 392 users with 3 active days reached 7 (74.0% drop-off).

[SIMULATED DATA RESULT] **Evidence:** Ladder (30-day horizon): 1→2 days 25.0% drop, 2→3 27.0%, 3→7 74.0%.

[ANALYST INTERPRETATION] **Interpretation:** Short-term returners rarely become weekly habits within a month.

Limitation: 7 active days in 30 is an analyst threshold; the ladder is sensitive to the window.

[PRODUCT HYPOTHESIS] **Product implication:** Investigate mid-term progression mechanics (missions, levels, streaks) as habit drivers.

FND-05 · Return rates depend strongly on calendar period, not on who the users are (Return · confidence High)

[SIMULATED DATA RESULT] **Observation:** D7 return rises from 26.3% for the 2026-04 first-active cohort to 56.4% for 2026-08; the daily active share of the registered base rose from 5.09% (May) to 11.34% (last 30 days).

[SIMULATED DATA RESULT] **Evidence:** Registration-month cohorts differ (χ^2 p=1.2e-16); source (p=0.54) and device (p=0.87) do not.

[ANALYST INTERPRETATION] **Interpretation:** Something time-dependent (simulation design, product changes, seasonality) drives return far more than user attributes.

Limitation: The dataset contains no release, campaign or notification calendar to explain the shift. TO BE VERIFIED.

[PRODUCT HYPOTHESIS] **Product implication:** Always control for cohort period when comparing groups; ask product/engineering what changed over time.

FND-06 · Fast activators return more, even after controlling for join month (Activation · confidence Medium)

[SIMULATED DATA RESULT] **Observation:** Users activating within 7 days have a 26.4pp higher fourth-week (D30) return than slower activators; within registration months the gap is 15.3pp.

[SIMULATED DATA RESULT] **Evidence:** Logistic model of D30 return: log(hours to first activity) standardised coefficient -0.524; CV AUC 0.765 (behaviour only 0.773, calendar only 0.595).

[ANALYST INTERPRETATION] **Interpretation:** Early activation is associated with later return; the association survives a calendar control.

Limitation: Association only – motivated users may both activate faster and return more (selection).

[PRODUCT HYPOTHESIS] **Product implication:** Test interventions that shorten time-to-first-activity and measure D30 return (EXP-01).

FND-07 · Two in five resolved tasks end in abandonment or failure, with no dominant failure reason (Tasks · confidence High)

[SIMULATED DATA RESULT] **Observation:** 3,571 of 9,198 resolved task records (38.8%) were abandoned (22.3%) or failed (16.5%).

[SIMULATED DATA RESULT] **Evidence:** Category completion rates range 59.8–62.5%; the five failure reasons each account for 701–732 records.

[ANALYST INTERPRETATION] **Interpretation:** Task friction is broad-based rather than concentrated in one category or failure mode.

Limitation: failure_reason is a coarse category; no error codes, third-party responses or UI context.

[PRODUCT HYPOTHESIS] **Product implication:** Instrument detailed failure context before redesigning specific task types.

FND-08 · Many task rewards sit in long-pending validation (Tasks · confidence Medium)

[SIMULATED DATA RESULT] **Observation:** 648 task records are Pending; 339 have been pending for more than 30 days at data end.

[SIMULATED DATA RESULT] **Evidence:** Pending share by category: 5.7–7.5%.

[ANALYST INTERPRETATION] **Interpretation:** Delayed reward confirmation may weaken the action→reward feedback loop (hypothesis).

Limitation: Whether users can see pending status, and validation SLAs, are not in the data. TO BE VERIFIED.

[PRODUCT HYPOTHESIS] **Product implication:** Review validation latency and pending-status visibility (EXP-04).

FND-09 · Ads, CAPTCHA and Missions each lose about half of attempts (Features · confidence Medium)

[SIMULATED DATA RESULT] **Observation:** Watch Ads skip rate 50.3%, CAPTCHA failure rate 49.7%, Missions abandonment rate 49.9% (event level).

[SIMULATED DATA RESULT] **Evidence:** Ads 1054 completed vs 1065 skipped; CAPTCHA 895 vs 883; Missions 702 vs 699.

[ANALYST INTERPRETATION] **Interpretation:** These are the three features where explicit negative outcomes are logged, and all show coin-flip completion.

Limitation: Other features log no failure/abandon events, so their completion rates are not comparable.

[PRODUCT HYPOTHESIS] **Product implication:** Investigate ad length/reward clarity, CAPTCHA difficulty and mission scope.

FND-10 · Viewing a mission or spin often does not lead to starting it (Features · confidence Low)

[SIMULATED DATA RESULT] **Observation:** 375 of 901 user×feature view pairs (41.6%) for Missions and Spin the Wheel never recorded a start of the same feature.

[SIMULATED DATA RESULT] **Evidence:** 369 pairs started without any recorded view.

[ANALYST INTERPRETATION] **Interpretation:** Exposure does not reliably convert to engagement on these surfaces.

Limitation: Views and starts are not linked in-session (DQ-035); this is a user-level signal only.

[PRODUCT HYPOTHESIS] **Product implication:** Test clearer start calls-to-action on mission and spin surfaces (EXP-05).

FND-11 · Sessions are short and longer sessions are not deeper (Sessions · confidence Medium)

[SIMULATED DATA RESULT] **Observation:** Median session 42s (P90 94s); duration↔events Spearman ρ = 0.027, duration↔meaningful ρ = 0.002.

[SIMULATED DATA RESULT] **Evidence:** Short (≤P25) sessions deliver 3.35 meaningful actions/min vs 0.87 for long (>P75).

[ANALYST INTERPRETATION] **Interpretation:** Session length is not a usable engagement proxy in the simulated data.

Limitation: 99.54% of completed tasks finish after their session 'ends' – session windows likely under-measure time in product.

[PRODUCT HYPOTHESIS] **Product implication:** Use completions and return frequency, not duration, as engagement KPIs; confirm session_end semantics.

FND-12 · First-day sessions are the densest; later sessions are lighter (Sessions · confidence Medium)

[SIMULATED DATA RESULT] **Observation:** First-active-day sessions average 7.64 events and 2.26 meaningful actions vs 4.17 and 1.2 on later days.

[SIMULATED DATA RESULT] **Evidence:** Mann-Whitney p = 3.3e-20; task completion is similar (61.17% vs 61.25%).

[ANALYST INTERPRETATION] **Interpretation:** New users explore intensively on day one; returning users make shorter, more targeted visits.

Limitation: Day-one density may partly reflect how the simulation seeded first sessions.

[PRODUCT HYPOTHESIS] **Product implication:** Design returning-user surfaces for quick, repeatable value actions.

FND-13 · Heavy first-day activity is associated with lower first-week return (Return · confidence Low)

[SIMULATED DATA RESULT] **Observation:** D7 return by first-day event-volume tercile: low 55.8%, mid 52.1%, high 33.9%.

[SIMULATED DATA RESULT] **Evidence:** In the D30 model, week-1 events have a negative standardised coefficient (-0.858) while week-1 sessions and active days are positive.

[ANALYST INTERPRETATION] **Interpretation:** Cramming many actions into few visits looks different from spreading activity across visits.

Limitation: Could reflect simulation design or exhaustion of daily earning limits; not causal.

[PRODUCT HYPOTHESIS] **Product implication:** Investigate whether pacing (daily caps, 'come back tomorrow' previews) relates to return.

FND-14 · Feature breadth is high and mostly mirrors activity volume (Features · confidence Medium)

[SIMULATED DATA RESULT] **Observation:** Active users touched a median of 13 of 15 features; 257 used all 15. Breadth vs active days Spearman ρ = 0.584.

[SIMULATED DATA RESULT] **Evidence:** Multi-feature (≥8 features completed): 265 users (26.9% of active users).

[ANALYST INTERPRETATION] **Interpretation:** Breadth is largely a by-product of more visits, so 'multi-feature' is not an independent engagement driver here.

Limitation: Simulated feature exposure may be more uniform than a real navigation structure.

[PRODUCT HYPOTHESIS] **Product implication:** Measure breadth per visit (not lifetime) when evaluating discovery changes.

FND-15 · Journeys show no strong sequential structure (Journeys · confidence Low)

[SIMULATED DATA RESULT] **Observation:** Feature-to-feature transition lifts range 0.79–1.27; session co-occurrence lifts 1.06–1.26; the most common single path covers only 1.17% of sessions.

[SIMULATED DATA RESULT] **Evidence:** After a view, the next event is a completion about as often as after any other stage (stage-transition matrix).

[ANALYST INTERPRETATION] **Interpretation:** Users do not follow a dominant path; no feature reliably leads to another.

Limitation: Event order inside sessions is unreliable (sequence numbers and timestamps disagree).

[PRODUCT HYPOTHESIS] **Product implication:** Fix event ordering instrumentation before investing in journey-based recommendations.

FND-16 · Streak behaviour does not look like streaks (Progression · confidence Medium)

[SIMULATED DATA RESULT] **Observation:** Only 23 of 533 streak claimers (4.3%) claimed on consecutive days; break events (919) ≈ claim events (922).

[SIMULATED DATA RESULT] **Evidence:** 207 users broke a streak without any recorded claim.

[ANALYST INTERPRETATION] **Interpretation:** Either the streak mechanic is not driving consecutive-day behaviour, or streak events are logged inconsistently.

Limitation: No streak length or state field exists. TO BE VERIFIED with engineering.

[PRODUCT HYPOTHESIS] **Product implication:** Add streak_started/streak_day_count events; then evaluate streak-saver ideas (EXP-06).

FND-17 · Levelling up is not followed by a change in activity (Progression · confidence Medium)

[SIMULATED DATA RESULT] **Observation:** 615 users (62.5% of active) levelled up. Around the first level-up: 1.2 active days in the prior 14 days vs 1.34 after (Wilcoxon p = 0.22, n = 187).

[SIMULATED DATA RESULT] **Evidence:** 615 levellers recorded 1266 level-up events (2.06 per leveller).

[ANALYST INTERPRETATION] **Interpretation:** Within the simulated data, progression milestones are not associated with a detectable activity lift.

Limitation: Level and XP balance are absent (level-ups counted from events); exact thresholds are simulated and documented VELOOP rules may change.

[PRODUCT HYPOTHESIS] **Product implication:** Investigate level-up feedback (reward visibility, next-goal messaging).

FND-18 · Device and acquisition source show no meaningful behavioural differences (Segments · confidence High)

[SIMULATED DATA RESULT] **Observation:** All Kruskal-Wallis/χ^2 tests across devices and sources have negligible effect sizes (max $\varepsilon^2 = 0.004$).

[SIMULATED DATA RESULT] **Evidence:** Device activation: Android 61.08%, iOS 61.61%, Windows 49.63%, macOS 47.83% (χ^2 p = 0.0344, Cramér's V = 0.0979).

[ANALYST INTERPRETATION] **Interpretation:** Observed behavioural differences are small; no device or source stands out on depth or return.

Limitation: Simulated acquisition sources do not represent real marketing performance; all traffic is 'Web'.

[PRODUCT HYPOTHESIS] **Product implication:** Do not prioritise device-specific fixes on this evidence; re-test with production data.

FND-19 · Activity is spread evenly across hours and days of the week (Time · confidence High)

[SIMULATED DATA RESULT] **Observation:** Hour-of-day session distribution is consistent with uniform (χ^2 p = 0.44); weekend vs weekday trend-adjusted DAU index 1.045 vs 0.974 (p = 0.11).

[SIMULATED DATA RESULT] **Evidence:** Weekdays hold 70.1% of events vs 71.4% of days.

[ANALYST INTERPRETATION] **Interpretation:** The simulated data does not support time-targeted notification or maintenance-window decisions.

Limitation: Timestamp timezone is undocumented.

[PRODUCT HYPOTHESIS] **Product implication:** Notification-timing experiments would need production data to be meaningful.

FND-20 · Pre-aggregated fields would materially change conclusions (Data quality · confidence High)

[SIMULATED DATA RESULT] **Observation:** events_in_session implies 10.44 events/session vs 4.689 from raw events; the milestone sheet reports 389 first logins vs 984 users with sessions.

[SIMULATED DATA RESULT] **Evidence:** See data_quality_impact.csv for 29 before/after comparisons.

[ANALYST INTERPRETATION] **Interpretation:** Using convenience fields without validation would mislead product decisions.

Limitation: Simulated data; production pipelines may differ.

[PRODUCT HYPOTHESIS] **Product implication:** Engineering: reconcile session counters, milestone generation and daily duration with raw logs.

20. Product Friction Register

Issue id	Behavioral signal	Affected users	evidence	Possible explanations	confidence	Product area	Investigation needed
BHV-001	Slow first session after registration	464	464 users waited > 7 days for a first session; median 6.1 days.	Verification delays; weak welcome prompt; registration without intent; app install friction.	Medium	Onboarding	Instrument registration →verification→first-open steps; interview recent registrants.
BHV-002	One-day users (no second active day)	158	158 of 984 active users had one active day; D1 return 14.5%.	No reason to return; no reminder; first-day earning limits reached; low perceived value.	High	Session continuation / notification strategy	Track notification sends/opens; test next-day hooks.
BHV-003	Task abandonment and failure	961	22.3% abandoned + 16.5% failed of resolved tasks.	Unclear instructions; technical failure; task unavailability; third-party dependency; timeouts.	High	Task UX	Capture failure detail (error code, step, partner response) per attempt.
BHV-004	Long-pending reward validation	481	648 pending records; 339 older than 30 days.	Third-party validation latency; manual review queues; missing status feedback.	Medium	Reward feedback	Measure validation latency; check whether pending status is visible to users.
BHV-005	Ad skipping	568	Skip rate 50.3% of ad outcomes.	Ad length; low reward per ad; ad relevance; playback errors.	Medium	Task UX / reward feedback	Log ad duration watched, reward offered and playback errors.
BHV-006	Mission abandonment	436	Abandonment rate 49.9% of mission outcomes.	Mission scope too large; unclear progress; competing priorities.	Medium	Progression	Add mission_progress events with step counts.
BHV-007	CAPTCHA failure	512	Failure rate 49.7% of CAPTCHA outcomes.	Difficulty; accessibility; automation filtering; input errors on mobile.	Medium	Task UX	Log failure type and attempt number; review accessibility.
BHV-008	Mission / spin viewed but not started	375	41.6% of view pairs never started.	Weak call-to-action; unclear reward; eligibility limits.	Low	Discoverability	Add feature_impression with CTA visibility; link view→start in-session.
BHV-009	Streak mechanics not producing consecutive claims	533	4.3% of claimers claimed on consecutive days; break/claim ratio 0.997.	Streak not visible; reset rules unclear; logging defect.	Low	Daily engagement	Confirm streak state logic with engineering; add streak_day_count.
BHV-010	Stall between 3rd and 7th active day	290	74.0% of 3-day users did not reach 7 days within 30 days.	No mid-term goals; repetitive activities; reward saturation.	Medium	Progression	Analyse feature mix of users who pass vs stall; test mid-term goals.
BHV-011	Sessions with no recorded feature activity	302	302 sessions (3.8%) have no events.	Navigation dead-ends; loading failures; tracking loss.	Low	Navigation	Add session_ended with end_reason and page_view events.

Possible explanations are hypotheses; none is confirmed by the dataset.

21. Product Opportunity Register

- reach: Affected simulated users: ≥30% = 3, 10–30% = 2, <10% = 1
- severity: Size of the behavioural loss at that step: large = 3, moderate = 2, small = 1
- confidence: From the friction register (data reliability of the signal): High 3, Medium 2, Low 1
- ease: Inverse implementation complexity: low complexity = 3, high = 1
- relevance: Closeness to the core earn → reward → return loop: core = 3, supporting = 2, peripheral = 1
- priority: Sum of five scores (5–15): High ≥ 12, Medium 9–11, Low ≤ 8

Opportunity id	finding	User problem	Product hypothesis	Proposed investigation	Affected population	Priority score	priority	measurement
OPP-03	Reduce task abandonment and failure	Two in five task attempts fail or are abandoned	Clearer instructions and retry paths may lift completion.	Failure-detail instrumentation, then EXP-03.	961	14	High	Task completion rate (resolved); retries per task
OPP-01	Shorten registration → first activity	Users register but wait days to start	A guided first-open flow and welcome prompt may shorten time-to-first-activity.	Funnel study of registration→first session; EXP-01.	464	13	High	14-day activation rate; median hours to first meaningful activity
OPP-02	Create a reason to return on day 2	Users do not come back after day one	Next-day reward previews / reminders may raise second-day return.	EXP-02 (next-day hook).	158	13	High	D1 and D7 return; share of one-day users
OPP-05	Improve ad completion	Half of ad attempts are skipped	Shorter/clearer-reward ads may reduce skipping.	Analyse skip vs ad metadata once logged.	568	11	Medium	Ad completion rate

Opportunity id	finding	User problem	Product hypothesis	Proposed investigation	Affected population	Priority score	priority	measurement
OPP-06	Right-size missions	Half of missions are abandoned	Smaller steps with progress feedback may raise completion.	Add mission_progress tracking; test step-based missions.	436	11	Medium	Mission completion rate
OPP-10	Close tracking gaps for empty sessions	Visits with no recorded activity	Better session/page instrumentation will separate UX dead-ends from tracking loss.	Instrumentation change.	302	11	Medium	Share of zero-event sessions
OPP-04	Make pending validation visible	Rewards wait in validation without feedback	Status tracking with expected time may sustain trust and return.	EXP-04 among users with pending tasks.	481	10	Medium	D7 return of users with pending tasks; support contacts
OPP-07	Convert mission/spin views into starts	Viewers often do not start	Stronger CTA and reward visibility on the surface.	EXP-05.	375	10	Medium	View→start conversion
OPP-08	Repair streak mechanics & tracking	Streak behaviour is inconsistent	Reliable streak state + streak saver may encourage consecutive days.	Engineering audit; then EXP-06.	533	9	Medium	Consecutive-day claim rate
OPP-09	Support mid-term progression (days 3–7)	Returners stall before forming a habit	Mid-term goals (weekly missions, level targets) may sustain activity.	Cohort analysis of passers vs stallers.	290	8	Low	Share of 3-day users reaching 7 days in 30

22. Dashboard

Live: https://veloop-data-analyst-task02.vercel.app/ · An interactive, client-facing dashboard (React + TypeScript + Vite + Tailwind + Recharts, folder *dashboard/*) built on aggregate, identifier-free data. It keeps business content only; technical material (data-quality register, anomaly queue, metric reconciliation, model details) stays in this report and the repository. Filters: acquisition source, device, user group, feature and activity dates. A simulated-data banner is shown on every page.

View	Content
Overview	Registered and active users, sessions, active days, activation, return, broad-usage and task-completion rates; daily active users; user journey; key findings
User Behaviour	Active users over time, visit frequency, session length, active days and the six user groups
Features	Feature usage vs completion, features users come back to, where attempts are lost (ads, CAPTCHA, missions), task results by category
Retention	Next-day, first-week, second-week and fourth-week return; weekly retention curve; first-week return by month joined
Insights & Actions	Where users drop off, the biggest problems found, and recommended actions with how to test them and what success looks like

Run locally: *cd dashboard && npm install && npm run dev*. Deploy: import the repository in Vercel with root directory *dashboard* (see README).

23. KPI Catalogue

Kpi id	category	Kpi name	formula	denominator	Data source	Current value simulated	limitation
KPI-01	User Activity	Registered Users	COUNT(user_id)	—	01_Users_1000	1000	Simulated population.
KPI-02	User Activity	Active Users	COUNT DISTINCT user_id in events	—	03_Feature_Events	984	Passive views count as activity.
KPI-03	User Activity	Active User Rate	Active Users / Registered Users	Registered users	01 + 03	98.4	Lifetime, not period-specific.
KPI-04	User Activity	Active Days per Active User	SUM(active days) / Active Users	Active users	03_Feature_Events	6.78	Tenure-dependent; median also reported (5).
KPI-05	User Activity	Sessions per User	Total Sessions / Registered Users	Registered users	02_Sessions	7.913	Includes zero-event sessions.
KPI-06	User Activity	Sessions per Active User	Total Sessions / Active Users	Active users	02_Sessions	8.042	Skewed – median 6.
KPI-07	User Activity	Events per Active User	Clean Events / Active Users	Active users	03_Feature_Events (clean)	36.27	Mean is sensitive to duplicates.
KPI-08	User Activity	Average DAU (last 30 days)	AVG(daily distinct users)	Days	03_Feature_Events	103.5	Grows with registered base.
KPI-09	User Activity	DAU/MAU Stickiness (last 30 days)	AVG(DAU_30d) / MAU_30d	Users active in last 30 days	03_Feature_Events	12.7	Single window at data end.
KPI-10	Session Behavior	Average Session Duration	SUM(duration) / Sessions	All sessions	02_Sessions	51.81	Windows end before tasks finish (DQ-022).
KPI-11	Session Behavior	Median Session Duration	MEDIAN(duration)	All sessions	02_Sessions	42.0	Not an engagement-quality measure.
KPI-12	Session Behavior	Events per Session	Clean Events / Valid Sessions	Valid sessions	02 + 03	4.689	events_in_session field is invalid.
KPI-13	Session Behavior	Features per Session	AVG(distinct features per session)	Valid sessions	03_Feature_Events	3.733	—
KPI-14	Session Behavior	Repeat Session Rate	Users(≥2 sessions) / Users(≥1)	Users with a session	02_Sessions	85.5	Right-censored.
KPI-15	Session Behavior	Zero-Event Session Rate	Sessions(0 events) / Sessions	All sessions	02 + 03	3.8	Cause unknown.
KPI-16	Session Behavior	Meaningful Session Rate	Sessions(meaningful) / Valid Sessions	Valid sessions	03_Feature_Events	69.3	Event completions only.
KPI-17	Activation	Activation Rate (14-day)	Activated / Users registered ≥14 days before data end	Eligible registrants	01 + 03 + 05	58.8	Analyst definition; window chosen by capture curve.
KPI-18	Activation	Early Activation Rate (7-day)	Activated7 / Eligible7	Users registered ≥7 days before data end	01 + 03 + 05	45.2	—

Kpi id	category	Kpi name	formula	denominator	Data source	Current value simulated	limitation
KPI-19	Activation	Time to First Activity	MEDIAN(first_event − registration)	Active users	01 + 03	150.5	Excludes never-active users.
KPI-20	Activation	Time to First Meaningful Activity	MEDIAN(first_meaningful − registration)	Users with a meaningful activity	01 + 03 + 05	175.3	Depends on meaningful-activity definition.
KPI-21	Activation	First Activity Rate	Active Users / Registered	Registered users	01 + 03	98.4	Same population as Active User Rate (lifetime).
KPI-22	Activation	First Session Rate	Users with session / Registered	Registered users	01 + 02	98.4	Session = login proxy.
KPI-23	Feature Engagement	Feature Adoption Rate	Feature users / Active Users	Active users	03_Feature_Events	68.5–85.0	Exposure-level; see stages table.
KPI-24	Feature Engagement	Feature Repeat Rate	Repeat users / Feature users	Feature users	03_Feature_Events	50.4–67.1	Volume-dependent.
KPI-25	Feature Engagement	Feature Completion Rate	Completions / (Completions + Skips/Fails/Abandons)	Outcome events	03_Feature_Events	Watch Ads 49.7 / CAPTCHA 50.3 / Missions 50.1	Only 3 features log negative outcomes.
KPI-26	Feature Engagement	Feature Abandonment Rate	Skips/Fails/Abandons / Outcome events	Outcome events	03_Feature_Events	Watch Ads 50.3 / CAPTCHA 49.7 / Missions 49.9	As above.
KPI-27	Feature Engagement	Multi-Feature Usage Rate	Multi-feature users / Active Users	Active users	03_Feature_Events	26.9	Tied to activity volume.
KPI-28	Feature Engagement	Task Completion Rate	Completed / (Completed+ Failed+Abandoned)	Resolved tasks	05_Task_Activity	61.2	Pending excluded.
KPI-29	Feature Engagement	Task Abandonment Rate	Abandoned / Resolved	Resolved tasks	05_Task_Activity	22.3	Coarse failure reasons.
KPI-30	Feature Engagement	Task Pending Rate	Pending / Tasks	All tasks	05_Task_Activity	6.6	No resolution timestamps.
KPI-31	Return	D1 Return Rate	Returned D1 / Eligible	Users first active ≥1 day before data end	03_Feature_Events	14.5	Calendar-dependent.
KPI-32	Return	D7 Return Rate	Returned D7 / Eligible	Users with full 7-day window	03_Feature_Events	47.3	Calendar-dependent.
KPI-33	Return	D14 Return Rate	Returned D14 / Eligible	Users with full 14-day window	03_Feature_Events	45.4	Calendar-dependent.
KPI-34	Return	D30 Return Rate	Returned D30 / Eligible	Users with full 30-day window	03_Feature_Events	41.8	Calendar-dependent.
KPI-35	Return	Repeat User Rate	Users(≥2 days) / Active Users	Active users	03_Feature_Events	83.9	Right-censored.
KPI-36	Engagement Depth	Feature Diversity	AVG(distinct features)	Active users	03_Feature_Events	11.81	Max 15.
KPI-37	Engagement Depth	Activity Frequency	MEDIAN(date_i − date_{i−1})	Active-day pairs	03_Feature_Events	4.0	Excludes one-day users.
KPI-38	Engagement Depth	Engagement Score (median)	Weighted sum of 6 scaled components	Active users	Derived	54.66	NOT an official VELOOP metric.
KPI-39	Engagement Depth	High-Frequency User Rate	HF users / Active Users	Active users	03_Feature_Events	26.3	Tenure-dependent.
KPI-40	Engagement Depth	One-Time User Rate	One-time / Active Users	Active users	03_Feature_Events	16.1	Right-censored.
KPI-41	Engagement Depth	Regular Engagement Rate	Regular users / Active Users	Active users	03_Feature_Events	62.5	Tenure-dependent.
KPI-42	Engagement Depth	Streak Consecutive-Claim Rate	Users(consecutive) / Claimers	Streak claimers	03_Feature_Events	4.3	No streak state field.
KPI-43	Engagement Depth	Level-Up Rate	Levellers / Active Users	Active users	03_Feature_Events	62.5	No level/XP values.
KPI-44	Data Quality	Session Counter Reconciliation Rate	Matching sessions / Sessions	All sessions	02 vs 03	4.8	Raw events assumed correct.

Definitions, units and interpretations for every KPI: outputs/tables/kpi_catalogue.csv and docs/kpi_catalogue.md.

24. Product Experiment Hypotheses

PROPOSED EXPERIMENTS. None of these was run; they are designs for future testing. Each design states the baseline (measured in the simulated data), the smallest change worth detecting, the sample size needed (two-sided two-proportion test, α = 0.05, power 80%) and the decision rule, so results can be judged objectively.

ID	Primary metric	Baseline %	MDE (pp)	Target %	Sample / arm	Unit	Days at simulated volume
EXP-01	14-day activation rate	58.8	5	63.8	1489	user	501
EXP-02	D1 return rate	14.5	3	17.5	2344	user	796
EXP-03	Task completion rate (resolved)	61.2	3	64.2	4078	task attempt (randomise by user)	149
EXP-04	D7 return of users with a pending task	46.3	5	51.3	1568	user	1089
EXP-05	Missions/Spin view→start conversion	58.4	5	63.4	1495	user×feature view	555
EXP-06	Consecutive-day streak claim rate	4.3	2	6.3	1969	user	1234
EXP-07	D7 return, top day-1 volume tercile	33.9	5	38.9	1453	user	1480
EXP-08	Ad completion rate	49.7	5	54.7	1566	ad attempt (randomise by user)	247
EXP-09	Mission completion rate	50.1	5	55.1	1565	mission attempt (randomise by user)	374
EXP-10	Share of 3-day users reaching 7 active days in 30 days	26	5	31	1279	user	1090

Duration uses the simulated data's tiny volume (≈6 new users per day), so most tests would take months; the sample size is the useful number – with production traffic, duration = 2 × sample size ÷ eligible users per day.

EXP-01 · PROPOSED EXPERIMENT

Field	Value
Hypothesis	Guiding newly registered users straight into a first earning activity will shorten time-to-first-meaningful-activity.
Problem (evidence)	Median 6.1 days to first session; 14-day activation 58.8%.
Target users	Newly registered users (all sources/devices)
Control	Current post-registration experience
Variant	Guided first-open: single recommended activity + welcome reward preview
Primary metric	14-day activation rate – baseline 58.8%, target 63.8% (+5 pp)
Sample size	1,489 users per arm; randomise by user
Secondary metrics	Hours to first meaningful activity; D7 return
Guardrail metrics	Verification completion rate; support contacts; unsubscribe rate
Analysis	Two-sided two-proportion z-test (χ^2), α = 0.05, power 80%, fixed horizon, intention-to-treat; attempt-level metrics use user-clustered standard errors.
Decision rule	Adopt the variant only if the primary metric improves by ≥ 5 pp with $p < 0.05$ and no guardrail worsens by more than its pre-agreed limit; otherwise keep the control.
Expected behavioural change	More users complete a first activity within 3 days
Required event tracking	user_registered, onboarding_started/completed, first_meaningful_activity

EXP-02 · PROPOSED EXPERIMENT

Field	Value
Hypothesis	A next-day hook shown at the end of the first active day will increase second-day return.
Problem (evidence)	25.0% of 30-day-horizon users never had a second active day; D1 return 14.5%.
Target users	Users on their first active day
Control	No next-day hook
Variant	End-of-day card previewing tomorrow's reward + opt-in reminder
Primary metric	D1 return rate – baseline 14.5%, target 17.5% (+3 pp)
Sample size	2,344 users per arm; randomise by user
Secondary metrics	D7 return; active days in first 14
Guardrail metrics	Notification opt-out; complaint rate; session error rate
Analysis	Two-sided two-proportion z-test (χ^2), $\alpha = 0.05$, power 80%, fixed horizon, intention-to-treat; attempt-level metrics use user-clustered standard errors.
Decision rule	Adopt the variant only if the primary metric improves by ≥ 3 pp with $p < 0.05$ and no guardrail worsens by more than its pre-agreed limit; otherwise keep the control.
Expected behavioural change	Higher share of users with a second active day
Required event tracking	notification_sent/opened, session_started, next_day_hook_shown

EXP-03 · PROPOSED EXPERIMENT

Field	Value
Hypothesis	Clear step instructions and a one-tap retry after failure will raise task completion.
Problem (evidence)	38.8% of resolved tasks abandoned or failed.
Target users	Users starting tasks in any category
Control	Current task flow
Variant	Inline instructions + retry button + failure explanation
Primary metric	Task completion rate (resolved) – baseline 61.2%, target 64.2% (+3 pp)
Sample size	4,078 task attempts per arm; randomise by user
Secondary metrics	Tasks completed per user; time start→complete
Guardrail metrics	Fraud/validation rejection rate; task error rate
Analysis	Two-sided two-proportion z-test (χ^2), $\alpha = 0.05$, power 80%, fixed horizon, intention-to-treat; attempt-level metrics use user-clustered standard errors.
Decision rule	Adopt the variant only if the primary metric improves by ≥ 3 pp with $p < 0.05$ and no guardrail worsens by more than its pre-agreed limit; otherwise keep the control.
Expected behavioural change	Fewer abandoned/failed attempts
Required event tracking	task_step_viewed, task_failed (error_code), task_retry

EXP-04 · PROPOSED EXPERIMENT

Field	Value
Hypothesis	Showing pending-validation status with an expected time will sustain return among users awaiting rewards.
Problem (evidence)	648 pending task records; many older than 30 days.
Target users	Users with ≥1 pending task
Control	No status visibility
Variant	Pending tracker with ETA and notification on credit
Primary metric	D7 return of users with pending tasks – baseline 46.3%, target 51.3% (+5 pp)
Sample size	1,568 users per arm; randomise by user
Secondary metrics	Support contacts; subsequent task starts
Guardrail metrics	Validation SLA breaches; reward reversal rate
Analysis	Two-sided two-proportion z-test (χ^2), $\alpha = 0.05$, power 80%, fixed horizon, intention-to-treat; attempt-level metrics use user-clustered standard errors.
Decision rule	Adopt the variant only if the primary metric improves by ≥ 5 pp with $p < 0.05$ and no guardrail worsens by more than its pre-agreed limit; otherwise keep the control.
Expected behavioural change	Users keep engaging while rewards validate
Required event tracking	reward_pending, reward_credited (latency), pending_status_viewed

EXP-05 · PROPOSED EXPERIMENT

Field	Value
Hypothesis	A prominent start call-to-action with the reward value on mission and spin surfaces will raise view→start conversion.
Problem (evidence)	41.6% of Missions/Spin view pairs never started.
Target users	Users viewing Missions or Spin the Wheel
Control	Current surface
Variant	Prominent CTA + reward value
Primary metric	View→start conversion (same session) – baseline 58.4%, target 63.4% (+5 pp)
Sample size	1,495 user×feature views per arm; randomise by user
Secondary metrics	Completion rate; meaningful actions per session
Guardrail metrics	Mission abandonment rate; spin error rate
Analysis	Two-sided two-proportion z-test (χ^2), $\alpha = 0.05$, power 80%, fixed horizon, intention-to-treat; attempt-level metrics use user-clustered standard errors.
Decision rule	Adopt the variant only if the primary metric improves by ≥ 5 pp with $p < 0.05$ and no guardrail worsens by more than its pre-agreed limit; otherwise keep the control.
Expected behavioural change	More views convert to starts
Required event tracking	feature_impression, feature_started with source_surface

EXP-06 · PROPOSED EXPERIMENT

Field	Value
Hypothesis	A one-time 'streak saver' will increase consecutive-day claims among streak users.
Problem (evidence)	Only 4.3% of claimers claim on consecutive days.
Target users	Users who claimed a streak at least once
Control	Current streak rules
Variant	One free streak save per week + visible streak counter
Primary metric	Consecutive-day claim rate – baseline 4.3%, target 6.3% (+2 pp)
Sample size	1,969 users per arm; randomise by user
Secondary metrics	Active days per week; D14 return
Guardrail metrics	Reward cost per active user; streak_broken complaints
Analysis	Two-sided two-proportion z-test (χ^2), $\alpha = 0.05$, power 80%, fixed horizon, intention-to-treat; attempt-level metrics use user-clustered standard errors.
Decision rule	Adopt the variant only if the primary metric improves by ≥ 2 pp with $p < 0.05$ and no guardrail worsens by more than its pre-agreed limit; otherwise keep the control.
Expected behavioural change	Longer streak runs
Required event tracking	streak_started, streak_day_count, streak_saved, streak_broken (prior_length)

EXP-07 · PROPOSED EXPERIMENT

Field	Value
Hypothesis	Previewing tomorrow's earning opportunities to heavy first-day users will improve first-week return.
Problem (evidence)	D7 return high-volume day-1 tercile 33.9% vs low 55.8%.
Target users	Users in the top tercile of day-1 events
Control	No pacing message
Variant	'Come back tomorrow' preview after daily limits
Primary metric	D7 return – baseline 33.9%, target 38.9% (+5 pp)
Sample size	1,453 users per arm; randomise by user
Secondary metrics	Active days in first 14; sessions/week
Guardrail metrics	Same-day meaningful actions (should not drop sharply)
Analysis	Two-sided two-proportion z-test (χ^2), $\alpha = 0.05$, power 80%, fixed horizon, intention-to-treat; attempt-level metrics use user-clustered standard errors.
Decision rule	Adopt the variant only if the primary metric improves by ≥ 5 pp with $p < 0.05$ and no guardrail worsens by more than its pre-agreed limit; otherwise keep the control.
Expected behavioural change	Heavier users spread activity over more days
Required event tracking	daily_limit_reached, next_day_preview_shown

EXP-08 · PROPOSED EXPERIMENT

Field	Value
Hypothesis	Showing the reward value before an ad and offering a shorter ad format will reduce ad skipping.
Problem (evidence)	Watch Ads skip rate 50.3% of ad outcomes.
Target users	Users who start a Watch Ads activity
Control	Current ad flow
Variant	Reward shown up front + shorter ad option
Primary metric	Ad completion rate (completed ÷ completed + skipped) – baseline 49.7%, target 54.7% (+5 pp)
Sample size	1,566 ad attempts per arm; randomise by user
Secondary metrics	Ads completed per user; meaningful actions per session
Guardrail metrics	Ad revenue per impression (if tracked); playback error rate
Analysis	Two-sided two-proportion z-test (χ^2), $\alpha = 0.05$, power 80%, fixed horizon, intention-to-treat; attempt-level metrics use user-clustered standard errors.
Decision rule	Adopt the variant only if the primary metric improves by ≥ 5 pp with $p < 0.05$ and no guardrail worsens by more than its pre-agreed limit; otherwise keep the control.
Expected behavioural change	Fewer skipped ads
Required event tracking	ad_started (duration, reward_offered), ad_skipped (seconds_watched), ad_error

EXP-09 · PROPOSED EXPERIMENT

Field	Value
Hypothesis	Splitting missions into smaller steps with visible progress will reduce mission abandonment.
Problem (evidence)	Missions abandonment rate 49.9% of mission outcomes.
Target users	Users who start a mission
Control	Current single-step missions
Variant	Step-based missions with a progress bar
Primary metric	Mission completion rate (completed ÷ completed + abandoned) – baseline 50.1%, target 55.1% (+5 pp)
Sample size	1,565 mission attempts per arm; randomise by user
Secondary metrics	Missions completed per user; D7 return
Guardrail metrics	Reward cost per completed mission; time to complete
Analysis	Two-sided two-proportion z-test (χ^2), $\alpha = 0.05$, power 80%, fixed horizon, intention-to-treat; attempt-level metrics use user-clustered standard errors.
Decision rule	Adopt the variant only if the primary metric improves by ≥ 5 pp with $p < 0.05$ and no guardrail worsens by more than its pre-agreed limit; otherwise keep the control.
Expected behavioural change	More missions finished
Required event tracking	mission_progress (step, steps_total), mission_completed, mission_abandoned (step)

EXP-10 · PROPOSED EXPERIMENT

Field	Value
Hypothesis	A weekly goal shown from the third active day will help returning users reach a seventh active day.
Problem (evidence)	74.0% of users with 3 active days did not reach 7 within their first 30 days.
Target users	Users on their 3rd active day
Control	No weekly goal
Variant	Weekly goal card (e.g. 'be active 4 days this week') with a bonus
Primary metric	Share reaching 7 active days within 30 days of first activity – baseline 26.0%, target 31.0% (+5 pp)
Sample size	1,279 users per arm; randomise by user
Secondary metrics	Active days per week; D30 return
Guardrail metrics	Bonus cost per active user; same-day meaningful actions
Analysis	Two-sided two-proportion z-test (χ^2), $\alpha = 0.05$, power 80%, fixed horizon, intention-to-treat; attempt-level metrics use user-clustered standard errors.
Decision rule	Adopt the variant only if the primary metric improves by ≥ 5 pp with $p < 0.05$ and no guardrail worsens by more than its pre-agreed limit; otherwise keep the control.
Expected behavioural change	More users form a weekly habit
Required event tracking	goal_shown, goal_progress, goal_completed

25. Event Tracking Gaps

All events below are PROPOSED analytics events – none exists in the simulated taxonomy.

Event name	trigger	Required properties	Why needed	Kpi supported
user_registered	Account created	user_id, ts, acquisition_source, campaign_id, device, platform	No registration event; registration time only in user master.	Activation rate, time to first activity
login	Successful authentication	user_id, ts, method, device, app_version	Login is currently inferred from sessions.	Active users, login frequency
session_started	App/web session begins	session_id, user_id, ts, entry_point, referrer, app_version	Session start is not an event; zero-event sessions cannot be explained.	Sessions/user, zero-event session rate
session_ended	Session ends	session_id, ts, end_reason (timeout/logout/crash), duration_s	Session windows end before tasks complete (DQ-022).	Session duration, exit analysis
onboarding_started / onboarding_completed	Onboarding step entry / completion	user_id, step, step_index, ts	onboarding_status is a snapshot with no timing.	Onboarding completion rate, activation
feature_impression	Feature tile/surface rendered in view	user_id, session_id, feature, surface, position	Exposure is only known for features that log 'viewed' events.	Feature exposure, view→start conversion
feature_started / feature_completed / feature_abandoned	Unified lifecycle per feature attempt	attempt_id, feature, ts, outcome, reward_value	Starts and completions cannot be linked (DQ-035).	Feature completion & abandonment rate
task_viewed	Task detail opened (independent of start)	task_id, category, position, ts	Every task record has a start, so view→start drop-off is invisible.	Task view→start conversion
task_failed (detailed)	Task attempt fails	task_id, error_code, step, partner_response, retry_count	Five coarse failure reasons are evenly distributed – not actionable.	Task failure rate by cause
reward_credited / reward_pending	Reward state changes	reward_id, task_id, amount, validation_latency_s, status	Pending tasks have no resolution timestamp.	Validation latency, reward feedback
first_meaningful_activity	First completed value action	user_id, feature, ts, hours_since_registration	Must be reconstructed from multiple tables.	Time to first meaningful activity
streak_started / streak_day_count / streak_broken	Streak state changes	user_id, streak_length, prior_length, ts	No streak length; breaks occur without recorded claims.	Streak maintenance rate
mission_progress	Mission step completed	mission_id, step, steps_total, ts	Missions are binary complete/abandon only.	Mission completion & step drop-off
engagement_milestone_reached	Milestone achieved (system-generated from raw events)	user_id, milestone, ts, source_event_id	Current milestone sheet is inconsistent with raw activity.	Lifecycle milestone rates
level_up (with values)	Level changes	user_id, old_level, new_level, xp_balance, ts	current_level and xp_balance are absent.	Level progression, XP velocity
notification_sent / notification_opened	Push/email sent and opened	user_id, campaign, channel, ts	Return cannot be linked to prompts.	Notification-driven return
referral_registered / referral_activated	Referee registers / activates	referrer_id, referee_id, ts	Brief examples absent; referral impact unmeasurable.	Referral conversion
withdrawal_page_viewed / withdrawal_requested	Withdrawal intent and request	user_id, amount, method, status	No withdrawal events in the simulated taxonomy.	Wallet→withdrawal funnel

26. Limitations — What This Dataset Cannot Tell Us

From the brief

- Simulated users are not actual VELOOP users; no statement here describes the real platform.
- No production behavioural data was used.
- No reliable causal inference: all relationships are associations within synthetic data.
- No guaranteed representation of the real user population (demographics, devices, sources).
- Third-party activity (ad networks, task partners, validation) may not be represented accurately.
- Exact production feature availability may change (VELOOP documentation notes features and mechanics can change).
- Simulated timestamps are not operational logs; timezone is undocumented.
- Simulated acquisition sources do not represent actual marketing performance.
- Simulated retention does not represent actual VELOOP retention.

Discovered during analysis

- No login, registration, onboarding, session or task events in the taxonomy (13 brief examples absent) – login is proxied by sessions.
- Within-session event order is unreliable: sequence_number duplicates and disagrees with timestamps; completions are not linkable to starts.
- Session windows (median 42 s) end before 99.54% of task completions – duration under-measures time in product.
- 06_Milestones contradicts raw activity and was not used as a source of truth.
- 04_Daily_Activity duration is inconsistent with sessions; only its event/feature counts reconcile.
- current_level and xp_balance are absent: level/XP progression is limited to event counts.
- platform = 'Web' for every record: platform analysis is not supported; device is used instead.
- Return metrics are right-censored for recent users; windows require full observability.
- Strong calendar-period effects confound any cohort comparison that does not stratify by join date.
- Workbook volumes (sessions, daily records, feature usage) are below the brief's recommended ranges (source discrepancy).
- Project duration differs between the workbook README (22–25 days) and the PDF (20–24 days): source discrepancy requiring confirmation.
- Snapshot user attributes (account_status, onboarding_status, verification_status) have no timestamps.

27. Final Product Analytics Recommendations

Framework (brief §45): Finding → Evidence → User impact → Product hypothesis → **how it will be tested** → **how success is measured** → decision rule. None claims a change will definitely improve retention; each must pass its test first.

REC-01 · Reduce task abandonment and failure (priority High)

Field	Value
Evidence (simulated)	22.3% abandoned + 16.5% failed of resolved tasks.
User impact	Two in five task attempts fail or are abandoned
Product hypothesis	Clearer instructions and retry paths may lift completion.
How to test	EXP-03: Current task flow vs Inline instructions + retry button + failure explanation; randomise by task attempt; 4,078 per arm.
How to measure	Primary: Task completion rate (resolved) – baseline 61.2% → target 64.2% (+3 pp). Guardrails: Fraud/validation rejection rate; task error rate.
Decision rule	Adopt the variant only if the primary metric improves by ≥ 3 pp with p < 0.05 and no guardrail worsens by more than its pre-agreed limit; otherwise keep the control.
Timeline	≈149 days at simulated volume – recompute with production traffic.

REC-02 · Shorten registration → first activity (priority High)

Field	Value
Evidence (simulated)	464 users waited > 7 days for a first session; median 6.1 days.
User impact	Users register but wait days to start
Product hypothesis	A guided first-open flow and welcome prompt may shorten time-to-first-activity.
How to test	EXP-01: Current post-registration experience vs Guided first-open: single recommended activity + welcome reward preview; randomise by user; 1,489 per arm.
How to measure	Primary: 14-day activation rate – baseline 58.8% → target 63.8% (+5 pp). Guardrails: Verification completion rate; support contacts; unsubscribe rate.
Decision rule	Adopt the variant only if the primary metric improves by ≥ 5 pp with p < 0.05 and no guardrail worsens by more than its pre-agreed limit; otherwise keep the control.
Timeline	≈501 days at simulated volume – recompute with production traffic.

REC-03 · Create a reason to return on day 2 (priority High)

Field	Value
Evidence (simulated)	158 of 984 active users had one active day; D1 return 14.5%.
User impact	Users do not come back after day one
Product hypothesis	Next-day reward previews / reminders may raise second-day return.
How to test	EXP-02: No next-day hook vs End-of-day card previewing tomorrow's reward + opt-in reminder; randomise by user; 2,344 per arm.
How to measure	Primary: D1 return rate – baseline 14.5% → target 17.5% (+3 pp). Guardrails: Notification opt-out; complaint rate; session error rate.
Decision rule	Adopt the variant only if the primary metric improves by ≥ 3 pp with p < 0.05 and no guardrail worsens by more than its pre-agreed limit; otherwise keep the control.
Timeline	≈796 days at simulated volume – recompute with production traffic.

REC-04 · Improve ad completion (priority Medium)

Field	Value
Evidence (simulated)	Skip rate 50.3% of ad outcomes.
User impact	Half of ad attempts are skipped
Product hypothesis	Shorter/clearer-reward ads may reduce skipping.
How to test	EXP-08: Current ad flow vs Reward shown up front + shorter ad option; randomise by ad attempt; 1,566 per arm.
How to measure	Primary: Ad completion rate (completed ÷ completed + skipped) – baseline 49.7% → target 54.7% (+5 pp). Guardrails: Ad revenue per impression (if tracked); playback error rate.
Decision rule	Adopt the variant only if the primary metric improves by ≥ 5 pp with p < 0.05 and no guardrail worsens by more than its pre-agreed limit; otherwise keep the control.
Timeline	≈247 days at simulated volume – recompute with production traffic.

REC-05 · Right-size missions (priority Medium)

Field	Value
Evidence (simulated)	Abandonment rate 49.9% of mission outcomes.
User impact	Half of missions are abandoned
Product hypothesis	Smaller steps with progress feedback may raise completion.
How to test	EXP-09: Current single-step missions vs Step-based missions with a progress bar; randomise by mission attempt; 1,565 per arm.
How to measure	Primary: Mission completion rate (completed ÷ completed + abandoned) – baseline 50.1% → target 55.1% (+5 pp). Guardrails: Reward cost per completed mission; time to complete.
Decision rule	Adopt the variant only if the primary metric improves by ≥ 5 pp with $p < 0.05$ and no guardrail worsens by more than its pre-agreed limit; otherwise keep the control.
Timeline	≈374 days at simulated volume – recompute with production traffic.

REC-06 · Close tracking gaps for empty sessions (priority Medium)

Field	Value
Evidence (simulated)	302 sessions (3.8%) have no events.
User impact	Visits with no recorded activity
Product hypothesis	Better session/page instrumentation will separate UX dead-ends from tracking loss.
How to test	Instrumentation change, validated by a before/after comparison of tracking coverage.
How to measure	Zero-event session rate and share of sessions with session_started/session_ended events.
Decision rule	Done when zero-event sessions are explained (end_reason logged) for ≥ 99% of sessions.
Timeline	One release cycle.

REC-07 · Make pending validation visible (priority Medium)

Field	Value
Evidence (simulated)	648 pending records; 339 older than 30 days.
User impact	Rewards wait in validation without feedback
Product hypothesis	Status tracking with expected time may sustain trust and return.
How to test	EXP-04: No status visibility vs Pending tracker with ETA and notification on credit; randomise by user; 1,568 per arm.
How to measure	Primary: D7 return of users with pending tasks – baseline 46.3% → target 51.3% (+5 pp). Guardrails: Validation SLA breaches; reward reversal rate.
Decision rule	Adopt the variant only if the primary metric improves by ≥ 5 pp with $p < 0.05$ and no guardrail worsens by more than its pre-agreed limit; otherwise keep the control.
Timeline	≈1,089 days at simulated volume – recompute with production traffic.

REC-08 · Convert mission/spin views into starts (priority Medium)

Field	Value
Evidence (simulated)	41.6% of view pairs never started.
User impact	Viewers often do not start
Product hypothesis	Stronger CTA and reward visibility on the surface.
How to test	EXP-05: Current surface vs Prominent CTA + reward value; randomise by user×feature view; 1,495 per arm.
How to measure	Primary: View→start conversion (same session) – baseline 58.4% → target 63.4% (+5 pp). Guardrails: Mission abandonment rate; spin error rate.
Decision rule	Adopt the variant only if the primary metric improves by ≥ 5 pp with $p < 0.05$ and no guardrail worsens by more than its pre-agreed limit; otherwise keep the control.
Timeline	≈555 days at simulated volume – recompute with production traffic.

REC-09 · Repair streak mechanics & tracking (priority Medium)

Field	Value
Evidence (simulated)	4.3% of claimers claimed on consecutive days; break/claim ratio 0.997.
User impact	Streak behaviour is inconsistent
Product hypothesis	Reliable streak state + streak saver may encourage consecutive days.
How to test	EXP-06: Current streak rules vs One free streak save per week + visible streak counter; randomise by user; 1,969 per arm.
How to measure	Primary: Consecutive-day claim rate – baseline 4.3% → target 6.3% (+2 pp). Guardrails: Reward cost per active user; streak_broken complaints.
Decision rule	Adopt the variant only if the primary metric improves by ≥ 2 pp with $p < 0.05$ and no guardrail worsens by more than its pre-agreed limit; otherwise keep the control.
Timeline	≈1,234 days at simulated volume – recompute with production traffic.

REC-10 · Support mid-term progression (days 3–7) (priority Low)

Field	Value
Evidence (simulated)	74.0% of 3-day users did not reach 7 days within 30 days.
User impact	Returners stall before forming a habit
Product hypothesis	Mid-term goals (weekly missions, level targets) may sustain activity.
How to test	EXP-10: No weekly goal vs Weekly goal card (e.g. 'be active 4 days this week') with a bonus; randomise by user; 1,279 per arm.
How to measure	Primary: Share reaching 7 active days within 30 days of first activity – baseline 26.0% → target 31.0% (+5 pp). Guardrails: Bonus cost per active user; same-day meaningful actions.
Decision rule	Adopt the variant only if the primary metric improves by ≥ 5 pp with $p < 0.05$ and no guardrail worsens by more than its pre-agreed limit; otherwise keep the control.
Timeline	≈1,090 days at simulated volume – recompute with production traffic.

REC-11 · Fix pre-aggregated fields that disagree with raw logs (priority High)

Field	Value
Evidence (simulated)	events_in_session overstated 2.2×; milestone sheet inconsistent; daily duration field 4.8× the session-derived value.
User impact	Analytics consumers could draw wrong conclusions.
Product hypothesis	Pipeline derivations are not built from the same raw source.
How to test	Automated reconciliation job comparing every aggregate with a recomputation from raw events.
How to measure	Number of mismatching rows per table per day.
Decision rule	Aggregates may be used for reporting only when mismatches = 0 for 7 consecutive days.
Timeline	One sprint.

REC-12 · Always control for cohort period in comparisons (priority High)

Field	Value
Evidence (simulated)	D7 return 26.3% (2026-04) vs 56.4% (2026-08).
User impact	Group comparisons can be confounded by when users joined.
Product hypothesis	Time-dependent factors dominate behaviour differences.
How to test	Report every group comparison both raw and stratified by registration month.
How to measure	Difference between raw and month-adjusted estimates.
Decision rule	Only act on differences that persist after cohort adjustment; experiments randomise within the same calendar period.
Timeline	Immediate (analysis standard).

Advanced analyses performed

analysis	method	Key result
Cohort analysis	First-active monthly & weekly cohorts, fixed 30-day horizon comparisons, Kruskal/χ^2 tests	Registration-month cohorts differ strongly (χ^2 p=1.2e-16); source and device cohorts do not.
Behavioural segmentation	Quantile-derived, hierarchical mutually exclusive segments	Six segments; High-Frequency = 259 users.
Sequence analysis	Timestamp-ordered feature paths, first-order transition matrix, lift vs independence	Transition lifts 0.79–1.27; stage transitions show no view→start→complete structure.
Feature affinity analysis	Session co-occurrence support and lift for all 105 feature pairs	Lift range 1.06–1.26 (weak affinities).
Activation analysis	Window sensitivity (1–30 days) with censoring-aware denominators and capture curve	14-day window selected; rate 58.8%.

analysis	method	Key result
Retention analysis	Window-based (D1/D7/D14/D30) and exact-day return, two anchors, weekly return curve	D7 47.3%, D30 41.8% (first-active anchor).
Anomaly detection	Tukey far-out fences, logical impossibility checks, duplicate-sequence detection	1319 anomaly records across 13 types.
Distribution analysis	Quartile-based duration bands, frequency distributions, Spearman duration↔depth	Duration↔events ρ = 0.027 (longer ≠ deeper).
Time-series analysis	Daily series, robust-z spike/drop detection on residual vs rolling median, trend-adjusted weekday index	2 flagged days; weekday/weekend p=0.11.
Correlation analysis	Spearman matrix of 13 user-level metrics	Strongest pair: Active days ↔ Sessions (ρ=0.954).
Confounding check (stratification)	Activation speed vs D30 return within registration-month strata	Raw difference 26.4pp vs month-adjusted 15.3pp.
Multivariable association model	Logistic regression, 5-fold CV ROC-AUC, calendar/source/device controls	CV AUC 0.765 (calendar only 0.595, behaviour only 0.773).
Funnel analysis	Activation funnel, 8-stage engagement funnel, drop-off matrix	Largest user-level drop-off: Third active day → seventh active day (first 30 days) (74.0%).
Device & acquisition segmentation	Group comparisons with Kruskal-Wallis / χ^2 and effect sizes	No metric shows a non-trivial effect size.
Engagement score validation	Sensitivity (9 scenarios), PCA weights, early-score predictive check	Min Spearman vs base 0.979; early-score↔later activity ρ=0.326.

Appendix A — Answers to the 22 required analytical questions (brief §43)

area	question	answer (simulated data)
User Lifecycle	1. How many simulated users become active?	984 of 1000 (98.4%) recorded ≥1 feature event; 984 completed a meaningful activity.
User Lifecycle	2. How quickly do users perform their first activity?	Median 6.3 days from registration to first feature event; 13.0% complete a meaningful activity within 1 day.
User Lifecycle	3. How many users never progress beyond initial activity?	158 active users had only one active day; 153 never had meaningful activity on a second day; 16 never became active.
User Lifecycle	4. How many users become repeat users?	826 users (83.9% of active) had ≥2 active days; 615 were active in ≥3 distinct weeks.
Sessions	5. How frequently do users return?	Median gap between active days 4 days; D1 14.5%, D7 47.3%, D14 45.4%, D30 41.8% (week windows).
Sessions	6. How long do sessions typically last?	Median 42s (mean 52s, P90 94s).
Sessions	7. How many users have only one session?	143 users (14.3% of registered).
Sessions	8. What does the session distribution look like?	Right-skewed: median 6 sessions per active user, top quartile ≥11; 4.689 events and 3.73 features per valid session.
Features	9. Which features receive the most usage?	Watch Ads (836 users, 3,126 events); Daily Reward (835 users, 2,498 events); Streak Rewards (832 users, 2,734 events)
Features	10. Which features have high repeat usage?	Missions 67.1%; Watch Ads 65.1%; Streak Rewards 63.6% of their users used them in ≥2 sessions.
Features	11. Which features have significant drop-off?	Watch Ads 50.3% abandonment (skip); CAPTCHA 49.7% failure; Missions 49.9% abandonment; 41.6% of Missions/Spin viewers never start.
Features	12. Which feature combinations occur most frequently?	Streak Rewards + Watch Ads (763 sessions, lift 1.116); all pair lifts are 1.06–1.26 (weak affinity).
Acquisition	13. Do behavioural patterns differ by acquisition source?	Only negligibly – no source shows a non-trivial effect size on activity, depth, activation or return.
Acquisition	14. Are some sources associated with deeper engagement?	'Other' and Telegram show the highest mean active days and scores, but differences are not statistically detectable (30-day active days p=0.36).
Devices	15. Are there meaningful behavioural differences across devices?	No meaningful differences; activation differs slightly (Android/iOS higher) with a negligible effect size.
Time	16. When are users most active?	No significant hour-of-day peak (χ^2 p=0.44); DAU grows with calendar time, peaking at 164 in September.
Time	17. Are there meaningful weekday/weekend differences?	No (trend-adjusted DAU index p=0.11).
Retention	18. Which cohorts return more frequently?	Later first-active cohorts (Aug–Sep) and fast activators; registration-month cohorts differ strongly (p=1.2e-16), source/device do not.
Retention	19. What behaviours are common among returning users?	Multi-session users more often claim daily rewards (67.78% vs 54.55%), streaks, spins and level-ups – associations reflecting more visits, not causes.
Product	20. Where are the largest behavioural friction points?	Second-day return, task abandonment/failure, 3→7 active-day stall, ad/CAPTCHA/mission negative outcomes, long-pending validation.
Product	21. Which areas deserve further product investigation?	Reduce task abandonment and failure; Shorten registration → first activity; Create a reason to return on day 2; Improve ad completion
Product	22. What additional data should VELOOP collect?	18 proposed events, led by login/session_started/session_ended, linked feature lifecycle events, detailed task_failed, reward_pending/credited and streak state.

Appendix B — Chart catalogue

Chart id	Chart name	Business question	dataset	calculation	population	limitation	Product relevance
C01	Daily Active User Trend	How has daily activity evolved; are there unusual days?	03_Feature_Events	Distinct users per calendar day; 7-day trailing mean; robust z of log DAU vs 15-day rolling median	All active users	No campaign/release calendar; spikes are not attributed to causes.	Capacity planning and anomaly monitoring.
C02	Session Frequency Distribution	How often do users return to open sessions?	02_Sessions	Count of sessions per user, binned	All 1,000 registered users	Lifetime counts are tenure-dependent.	Sizing re-engagement audiences.
C03	Session Duration Distribution	How long do sessions typically last?	02_Sessions	Histogram (5s bins), quartile markers	All 7,913 sessions	Session windows end before task completions (DQ-022).	Do not use duration as an engagement KPI.
C04	Activation Funnel	What happens after a user joins?	01, 02, 03, 05	Cumulative user counts per stage / registered users	1,000 registered users	Login proxied by sessions; completions not linked to starts.	Focus activation work on return and first-session delay.
C05	Feature Adoption Comparison	Which features reach the most users, and how many complete them?	03_Feature_Events	Users with any event / active users; users with ≥1 'complete' event / active users	984 active users	Exposure only known when a feature logs events.	Feature prioritisation and discovery work.
C06	Feature Repeat-Use Comparison	Which features do users come back to?	03_Feature_Events	Users with feature in ≥2 sessions / feature users	Feature users	Heavier users have more chances to repeat any feature.	Identify sticky vs one-off features.
C07	User Engagement Segments	What behavioural groups exist and how big are they?	Derived users table	Hierarchical rules (engagement_segments.csv)	1,000 users	Thresholds are quantiles of this dataset.	Targeting and sizing.
C08	Acquisition Source Behaviour Comparison	Do behavioural patterns differ by acquisition source?	Users + sessions + events + tasks	Per-group rates; Kruskal-Wallis / χ^2 tests	1,000 users	Simulated sources/devices; all platform = Web.	Avoid ranking by registration volume alone.
C09	Device Behaviour Comparison	Do behavioural patterns differ by device?	Users + sessions + events + tasks	Per-group rates; Kruskal-Wallis / χ^2 tests	1,000 users	Simulated sources/devices; all platform = Web.	Avoid ranking by registration volume alone.
C10	Retention / Cohort Chart	Which cohorts return more frequently?	03_Feature_Events	Distinct active users per cohort-week / cohort size	984 active users	Right-censoring; small early cohorts.	Retention monitoring; control for cohort period.
C11	Feature Journey Visualization	What does a user's activity journey look like?	03_Feature_Events	Collapsed consecutive features; row-normalised transition counts	7,611 sessions	In-session order unreliable (DQ-033/034); no causal reading.	Journey-based recommendation needs better ordering data.
C12	Drop-Off Visualization	Where do users stop progressing?	01, 02, 03, 05	(entering − continuing) / entering	Varies by stage	Mixed units (users, records, user×feature).	Prioritise investigation areas.
C13	Time-of-Day Activity	When are users most active?	02_Sessions	Sessions per hour; χ^2 uniformity test	All sessions	Timezone unknown.	No evidence for time-targeted notifications from this data.
C14	Behavioural Anomaly Chart	Which users behave unusually?	Derived users + anomaly register	Tukey Q3+3·IQR fences; streak logic check	984 active users	Generic statistical thresholds only.	Operations review queue.
C15	Task Outcome by Category	Where does task friction concentrate?	05_Task_Activity	Status share of records	9,846 task records	Coarse failure reasons.	Task UX prioritisation.
C16	Calendar-Period Effect on Return	Is return behaviour stable over time?	03_Feature_Events	D7 returned / D7 eligible per first-active month	910 users	Cause unknown (TO BE VERIFIED).	Stratify analyses by cohort period.
C17	Engagement Score Distribution	How is engagement depth distributed?	Derived users table	Weighted sum ×100 (engagement_score_components.csv)	984 active users	Analyst weights; tenure bias.	Prioritisation of engagement cohorts.
C18	Data Quality Impact	Do data-quality problems change the conclusions?	All sheets	(after − before) / before	Metric-specific	Simulated stress tests are illustrative.	Engineering reconciliation priorities.

Appendix C — Source register

- Task 02 brief (Data_Analyst_02.pdf) – formal requirement document.
- VELOOP_Simulated_User_Behavior_1000.xlsx – simulated dataset (10 sheets).

- Product-context baseline cited by the brief (not independently accessed for this analysis): VELOOP Rewards Help Center, About Us, Privacy Policy, Level System and Disclaimer pages. Statements tagged DOCUMENTED PRODUCT FACT are limited to what the brief itself attributes to these sources.

Appendix D — Metric reconciliation

Numbers that look alike but differ by definition or denominator. Every pair below was recomputed by the pipeline; differences are explained, not errors.

ID	Metrics compared	Value A	Value B	Definition difference	Why they differ
REC-M01	Registered users vs active users	1,000	984	All accounts vs accounts with ≥1 feature event.	16 registered users never opened a session or used a feature.
REC-M02	One-session users vs one-time (1 active day) users	143	158	≥1 session counted per session vs per calendar day.	16 users had 2+ sessions, all on the same day: one-time by days, not by sessions.
REC-M03	Repeat-session rate vs repeat-user rate	85.5% (841/984)	83.9% (826/984)	≥2 sessions vs ≥2 distinct active days.	Same reason as above (several sessions on one day).
REC-M04	Repeat-user rate (KPI) vs 'Returned' (funnel)	826 users	826 users	Identical rule: ≥2 active days.	Equal by construction – shown to confirm consistency.
REC-M05	Returned vs Established	826	733	≥2 active days vs meaningful activity on ≥3 days.	Established is the stricter next funnel step.
REC-M06	Multi-feature: KPI (standalone) vs funnel (cumulative)	265 (26.9% of active)	235	Meets the breadth rule vs meets it AND every earlier funnel stage.	The funnel only counts multi-feature users who also returned.
REC-M07	Regular engagement: standalone vs funnel (cumulative)	615 (62.5% of active)	211	≥3 active weeks vs ≥3 active weeks AND multi-feature AND returned.	Cumulative counts shrink at every stage; standalone counts do not.
REC-M08	Completed a meaningful activity (ever) vs activation rate	984 users (98.4%)	58.8% (530/902)	Any time in the window vs within 14 days of registration, eligible users only.	Most users eventually complete something, but many only after the activation window.
REC-M09	Repeat-user rate (ever) vs D7 return	83.9%	47.3% (430/910)	Any second active day, ever vs a return within days 1–7 after the first active day.	Returning eventually is far more common than returning within the first week.
REC-M10	One-time users (all time) vs one-day users in first 30 days	16.1% (158/984)	25.0% (179/716)	All active users, lifetime vs users with ≥30 observable days, first 30 days only.	The 30-day version removes right-censoring and is the fairer comparison.
REC-M11	Sessions per user: registered vs active denominator	7.913	8.042	Sessions ÷ 1,000 registered vs ÷ active users.	Denominator stated wherever the metric appears.
REC-M12	Events per session: session-sheet field vs raw events	10.44	4.689	events_in_session field vs clean raw events ÷ valid sessions.	The field does not reconcile with raw events (DQ-016) and is not used anywhere.
REC-M13	D7 return: first-active anchor vs registration anchor	47.3%	50.3%	Days 1–7 after first active day vs after registration date.	Primary metric uses the first active date; the registration version is a labelled comparison only.

Registration-anchored return (comparison only)

Window	Days after registration	Eligible users	Returned users	Return rate %
D1	1	994	178	17.9
D7	1–7	953	479	50.3
D14	8–14	904	400	44.2
D30	24–30	816	333	40.8

Appendix E — Independent verification

- Source code: analysis/ (23 pipeline steps, one per analysis area) and dashboard/ – shared with this report.
- Dataset: data/anonymised/ contains every sheet of the supplied workbook as CSV with personal fields (name, email, age, gender, city, state, referral code) removed; simulated user IDs are kept so joins can be checked.
- Methodology: docs/methodology.md (every definition), docs/verification_guide.md (step-by-step checks).
- verify_headline_metrics.py recomputes the headline numbers directly from data/anonymised/ with plain pandas, without using the pipeline code, and compares them with the published outputs.
- Automated tests: tests/ (Python) and dashboard/src/lib/*.test.ts check arithmetic, denominators, funnel consistency and that the dashboard reproduces the pipeline numbers for every filter value.